

Formulary for vibrational energy harvesting

Available average power:

$$P_{el}(\omega, c_{harv}) = \frac{1}{2} \cdot c_{harv} \cdot \frac{\omega^4}{(\omega^2 - \omega_0^2)^2 + \left(\frac{c_{el} + c_{dis}}{m\omega_0}\right)^2 \omega^2 \omega_0^2} a_{vib}^2$$

At resonance:

$$P_{el}(c_{harv}) = \frac{1}{2} \cdot \frac{c_{harv}}{(c_{harv} + c_{dis})^2} \cdot m^2 \cdot a_{vib}^2$$

With matched generator:

$$P_{el,max} = \frac{1}{8} \cdot \frac{1}{c_{dis}} \cdot m^2 \cdot a_{vib}^2$$

$$P_{el,max} = \frac{1}{8} \cdot \frac{1}{c_{dis}} \cdot m^2 \cdot \omega_0^4 \cdot x_{vib}^2$$

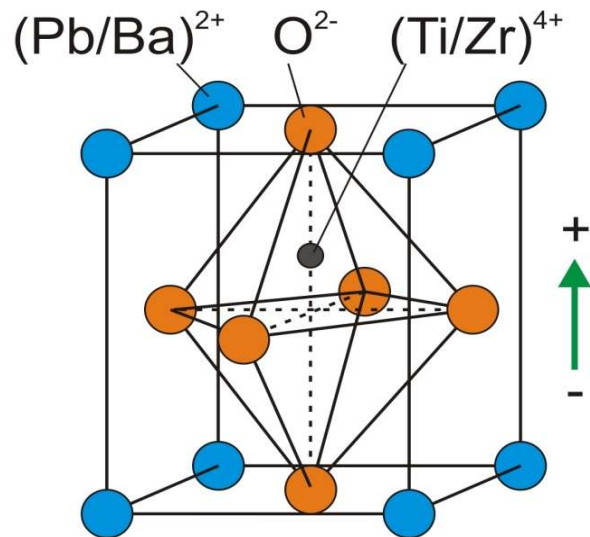
Content:

- Mathematical treatment of vibrations
- How to maximize the vibrational amplitude
- Condition for maximum power output
- How to find the right harvester design
- Piezoelectric harvesters
- Electromagnetic harvesters
- Electrostatic harvesters

Piezoelectric Energy Harvesters – Effect I

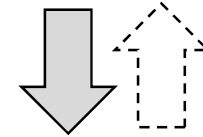
Principle: Direct piezoelectric effect

Charge displacement in a non-centrosymmetric crystal lattice, obtained via a mechanical deformation of the piezoelectric material

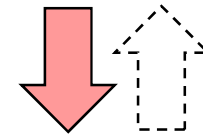


Perovskite crystal structure of standard piezoceramic material (here PbZrTi or PZT and BaZrTi)

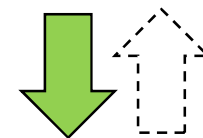
mechanical energy



mechanical deformation



charge displacement



electrical energy

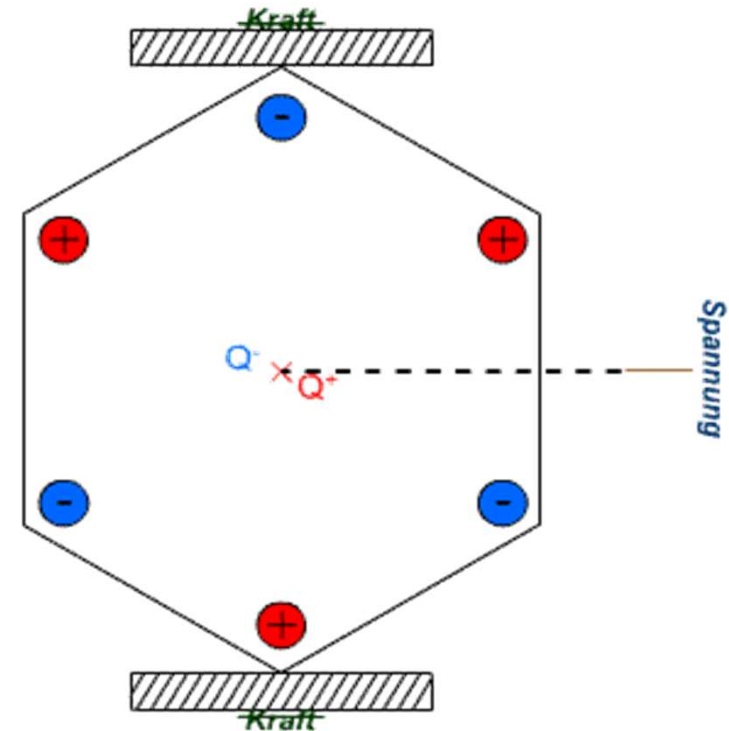
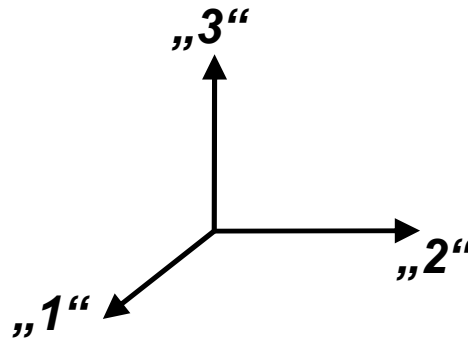
Piezoelectric Energy Harvesters – Effect II

„d₃₃-Mode“

- vertical force („3“)
- vertical displacement
- vertical charge dislocation

„d₃₁-Mode“

- **horizontal** force („1“ or „2“)
- horizontal and vertical displacement
- vertical charge dislocation



Piezoelectric Energy Harvesters – Effect III

Material parameters

d : piezoelectric strain coefficient $\left[\frac{m}{V} \right]$

e : dielectric constant $\left[\frac{As}{Vm} \right]$

Y : Young's modulus $\left[\frac{N}{m^2} \right]$

Variables

σ : mechanical stress $\left[\frac{N}{m^2} \right]$

ε : mechanical strain

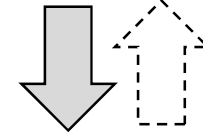
E : electrical field strength $\left[\frac{V}{m} \right]$

D : electrical displacement $\left[\frac{As}{m^2} \right]$

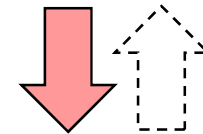
$$\underbrace{\varepsilon}_{\text{Hooke's Law}} = \underbrace{\frac{\sigma}{Y}}_{\text{Hooke's Law}} + \underbrace{d \cdot E}_{\text{piezoelectric coupling}}$$

$$\underbrace{D}_{\text{dielectric equation}} = \underbrace{e \cdot E}_{\text{dielectric equation}} + \underbrace{d \cdot \sigma}_{\text{piezoelectric coupling}}$$

mechanical energy



mechanical deformation



charge displacement



electrical energy

Piezoelectric Energy Harvesters – Effect IV

	units	PZT	PVDF	PZN-PT
strain coefficient d_{31}	10^{-12} m/V	320	20	950
strain coefficient d_{33}	10^{-12} m/V	650	30	2000
coupling coefficient k_{31}	-----	0,44	0,11	0,5
coupling coefficient k_{33}	-----	0,75	0,16	0,91
rel. dielectric constant	ϵ_r / ϵ_0	3800	12	4500
Young's modulus Y	10^{10} N/m ²	5	0,3	0,83
tensile strength σ_t	10^7 N/m ²	2	5,2	8,3

*S. Roundy et al.,
„Energy Scavengers
for Wireless Sensor
Networks“, Kluwer,
2004*

Piezoelectric coupling coefficient

Defines the ability to convert mechanical into electrical energy or vice versa, here:

$$k = \sqrt{\frac{\text{electrical energy stored}}{\text{mechanical energy applied}}}$$

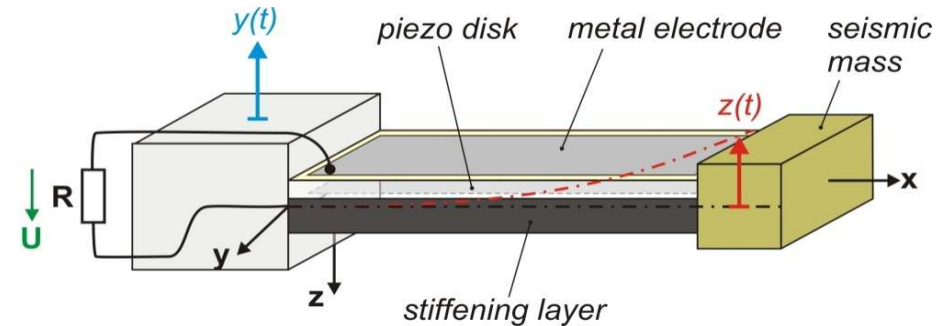
$$k = d \cdot \sqrt{\frac{Y}{e}}$$

*Y= Young's modulus
(formerly known as E)
Renamed for
clarification and
simplicity*

Piezoelectric Energy Harvesters – Working principle

Layer composition

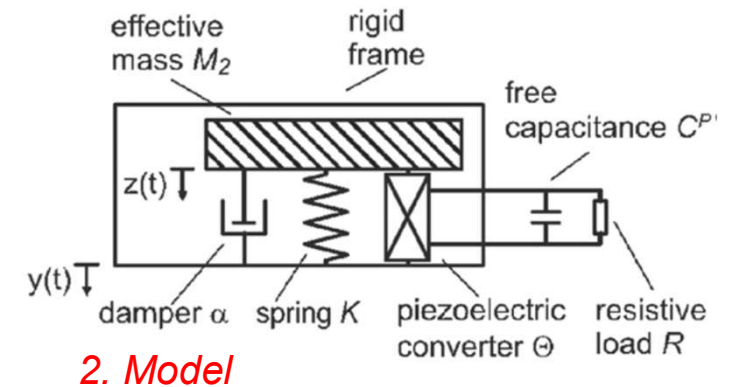
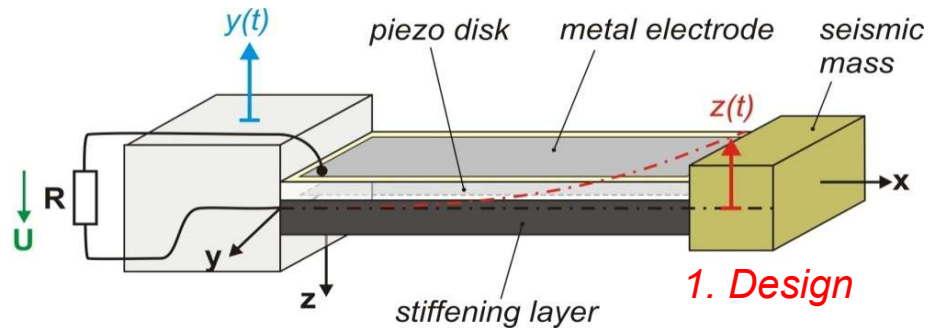
- generator as double layer or triple layer beam (1...2 piezo + 1 stiffening layer)
- „double nature“:
generator is also the spring of the seismic mass



Design challenges

- optimization of stress distribution
- tunable resonance frequency
- smart system integration
- ➔ higher output power
- ➔ broader application range, more power
- ➔ cheaper, easier fabrication

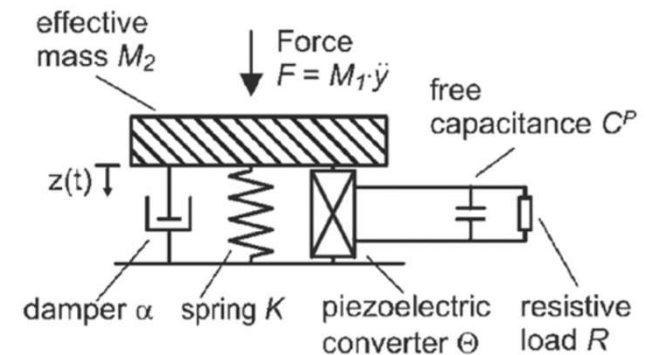
Piezoelectric Energy Harvesters – Modelling I



$$-m \cdot \ddot{x}(t) - c \cdot \dot{x}(t) - K \cdot x(t) = F_0 \cdot \cos \omega t$$

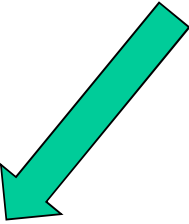
3. Equations

$$-\ddot{x}(t) - \frac{1}{m} \left(c_{dis} + \frac{mk^2 RC \omega_0^2}{(RC \omega_0)^2 + (1 - k^2)^2} \right) \cdot \dot{x}(t) - \omega_0^2 \left(1 + \frac{k^2(1 - k^2)}{(RC \omega_0)^2 + (1 - k^2)^2} \right) \cdot x(t) = \frac{F_0}{m} \cdot \cos \omega t$$



Piezoelectric Energy Harvesters – Modelling II

$$-\ddot{x}(t) - \frac{1}{m} \left(c_{dis} + \frac{mk^2 RC \omega_0^2}{(RC \omega_0)^2 + (1 - k^2)^2} \right) \cdot \dot{x}(t) - \omega_0^2 \left(1 + \frac{k^2(1 - k^2)}{(RC \omega_0)^2 + (1 - k^2)^2} \right) \cdot x(t) = \frac{F_0}{m} \cdot \cos \omega t$$


$$\zeta \approx \frac{c_{dis}}{2m\omega_0} + \frac{\frac{1}{2}k^2 RC \omega_0}{(RC \omega_0)^2 + (1 - k^2)^2}$$

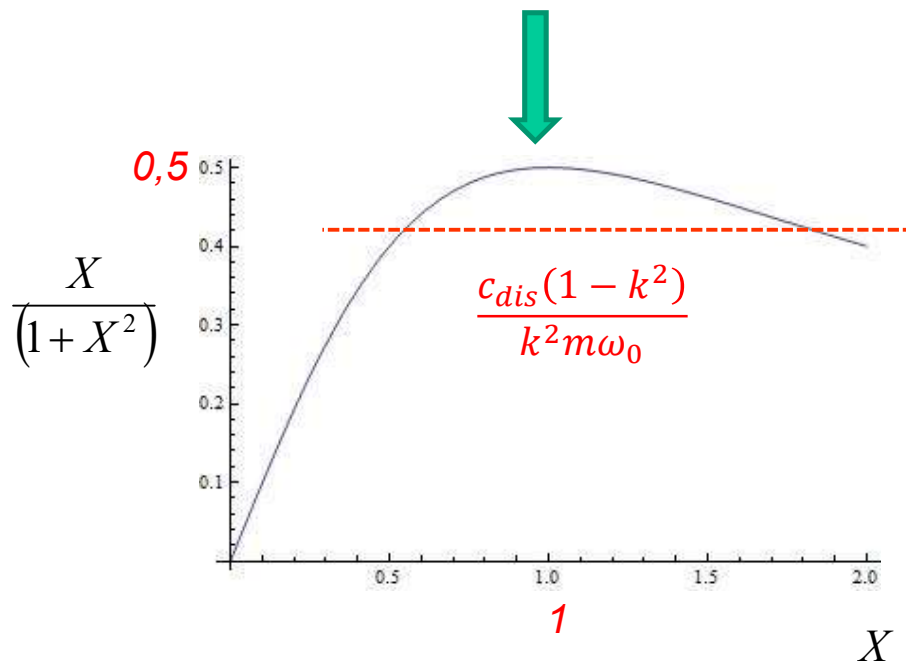
4. Identification of damping ratio

5. Load matching

$$\frac{c_{dis}}{2m\omega_0} = \frac{\frac{1}{2}k^2 RC \omega_0}{(RC \omega_0)^2 + (1 - k^2)^2}$$

$$\frac{c_{dis}(1 - k^2)}{k^2 m \omega_0} = \frac{X}{X^2 + 1}, X = \frac{RC \omega_0}{1 - k^2}$$

Optimum load



$$\frac{c_{dis}(1-k^2)}{k^2 m \omega_0} = \frac{X}{X^2+1}, X = \frac{RC\omega_0}{1-k^2}$$

*Low coupling:
Maximize the electrical damping*

$$\frac{c_{dis}(1-k^2)}{k^2 m \omega_0} > \frac{1}{2}; \quad X_{opt} = 1 \Rightarrow R_{opt} = \frac{1}{C\omega_0}$$

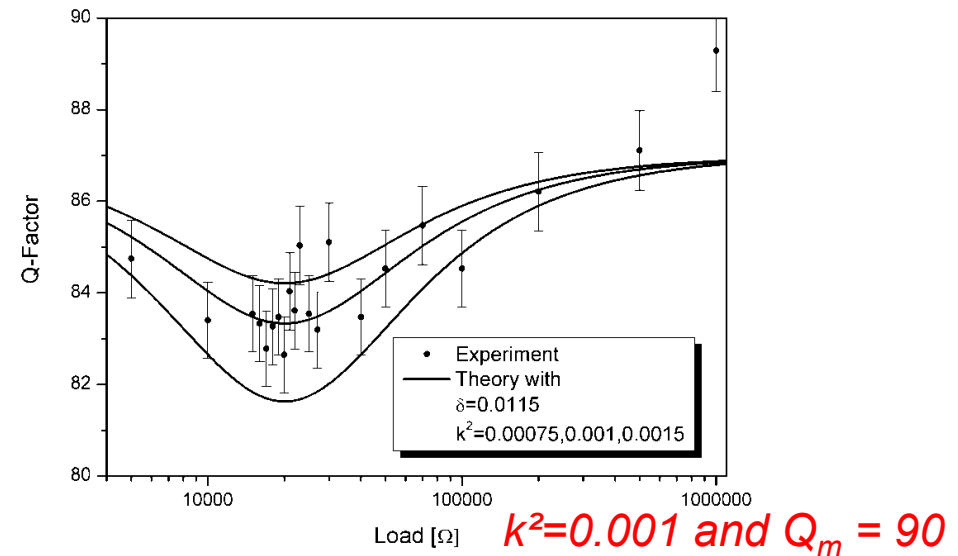
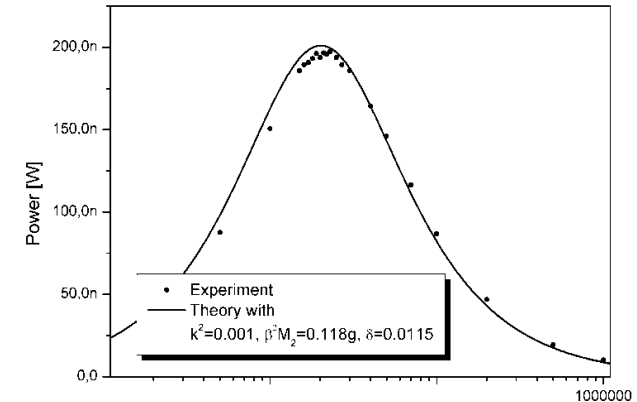
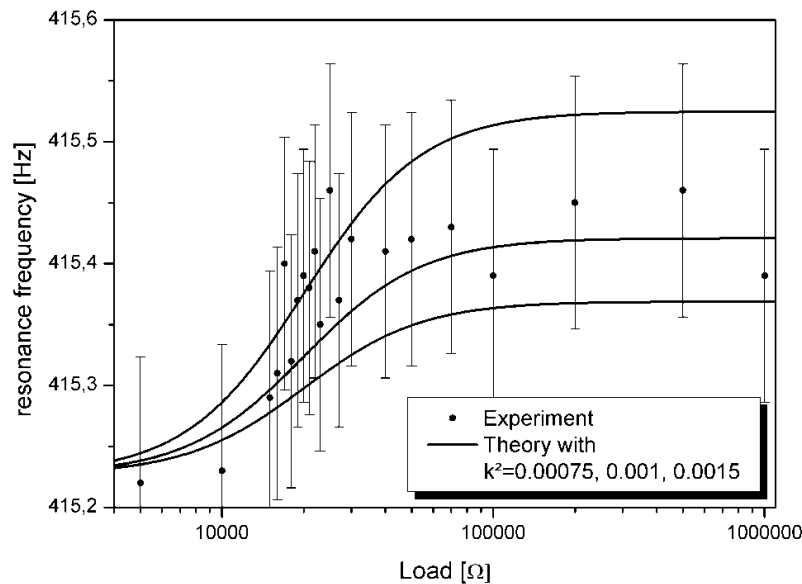
*High coupling:
Two optimum loads exist!*

$$\frac{c_{dis}(1-k^2)}{k^2 m \omega_0} < \frac{1}{2}; \quad X_{opt1} < 1 < X_{opt2}$$

Piezoelectric Energy Harvesters – Examples I

Low coupling harvester

- Slight resonance frequency change with the electrical load
- Small influence of the load on the Q-factor
- One optimum load at $R_{opt} = \frac{1}{C^S \omega_0}$



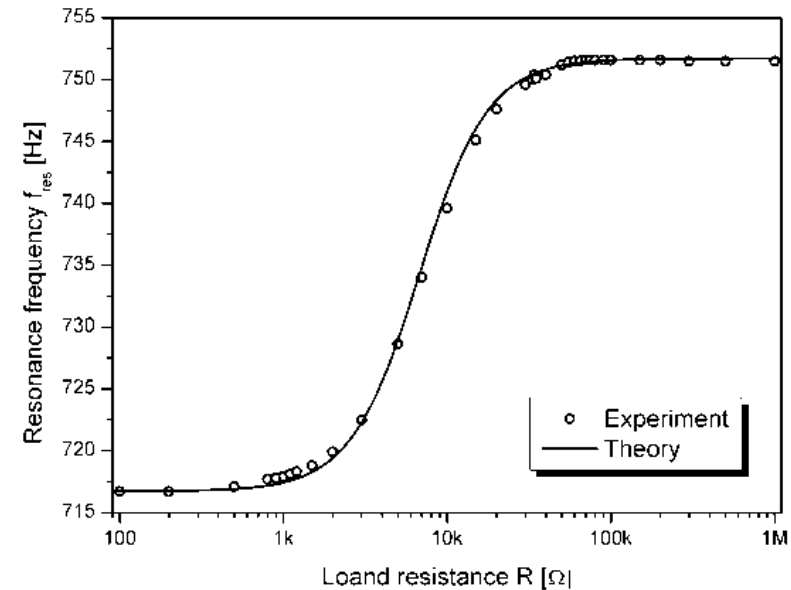
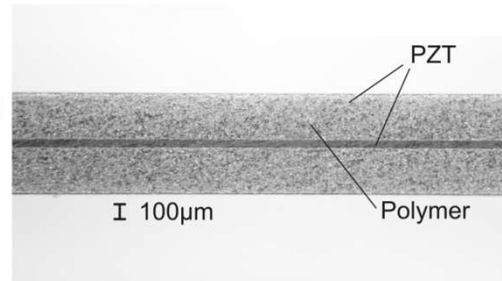
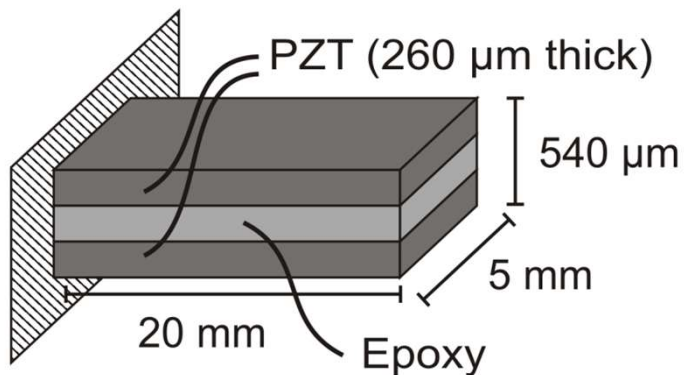
Piezoelectric Energy Harvesters – Examples II

High coupling Harvester

Significant change in resonance frequency

- *High influence of the load on the Q-factor*
- *Two optimum loads at*

$$R_{opt,1} \approx \frac{\omega_0}{C^S} \cdot \frac{2\zeta_m}{k^2}, \quad R_{opt,2} \approx \frac{\omega_0}{C^S} \cdot \frac{k^2}{2\zeta_m}$$



$$Q_m = 90 \text{ and } k^2 = 0.091$$

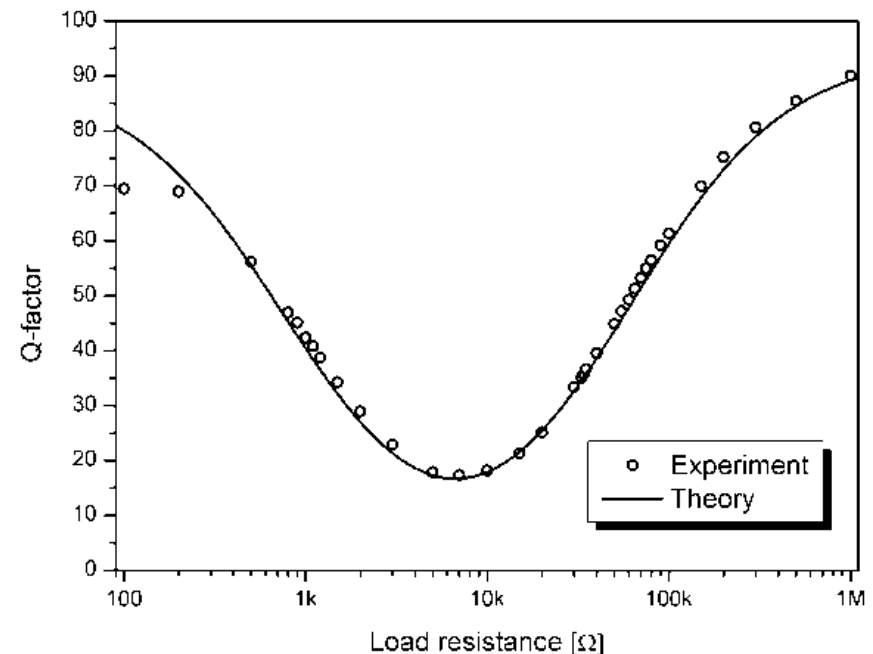
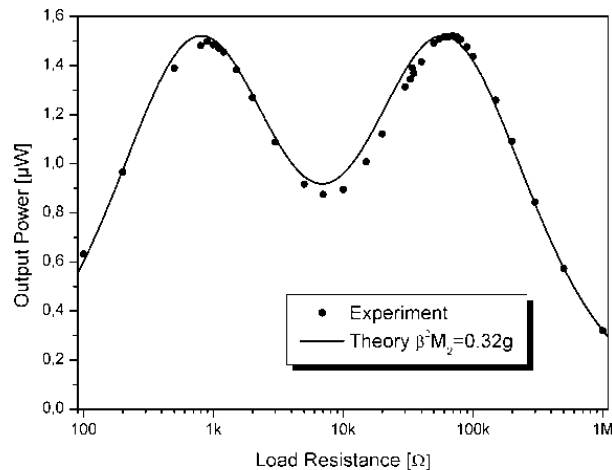
Piezoelectric Energy Harvesters – Examples III

High coupling Harvester

- Significant change in resonance frequency
- High influence of the load on the Q-factor
- Two Optimum loads at

$$Q_m=90 \text{ and } k^2=0.091$$

$$R_{opt,1} \approx \frac{\omega_0}{C^S} \cdot \frac{2\zeta_m}{k^2}, \quad R_{opt,2} \approx \frac{\omega_0}{C^S} \cdot \frac{k^2}{2\zeta_m}$$



Piezoelectric Energy Harvesters – Examples IV

$$q = d \cdot \sigma \quad \text{applies locally}$$
$$Q = C \cdot U \quad \text{applies globally}$$

Inhomogeneous stress distribution

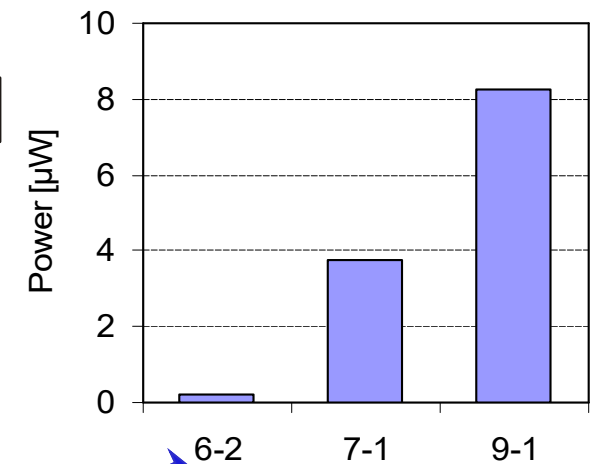
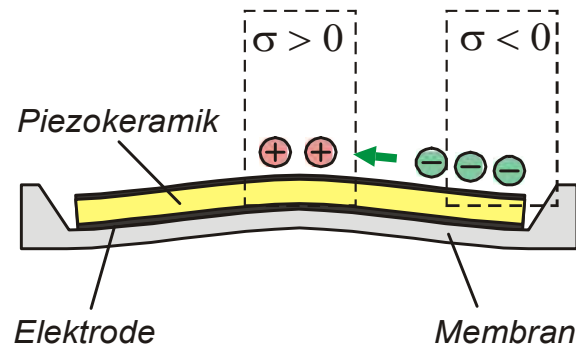
- inhomogeneous charge generation
- internal short circuits via large area electrodes
- internal self-blocking

➔ **energy losses**

Potential solutions

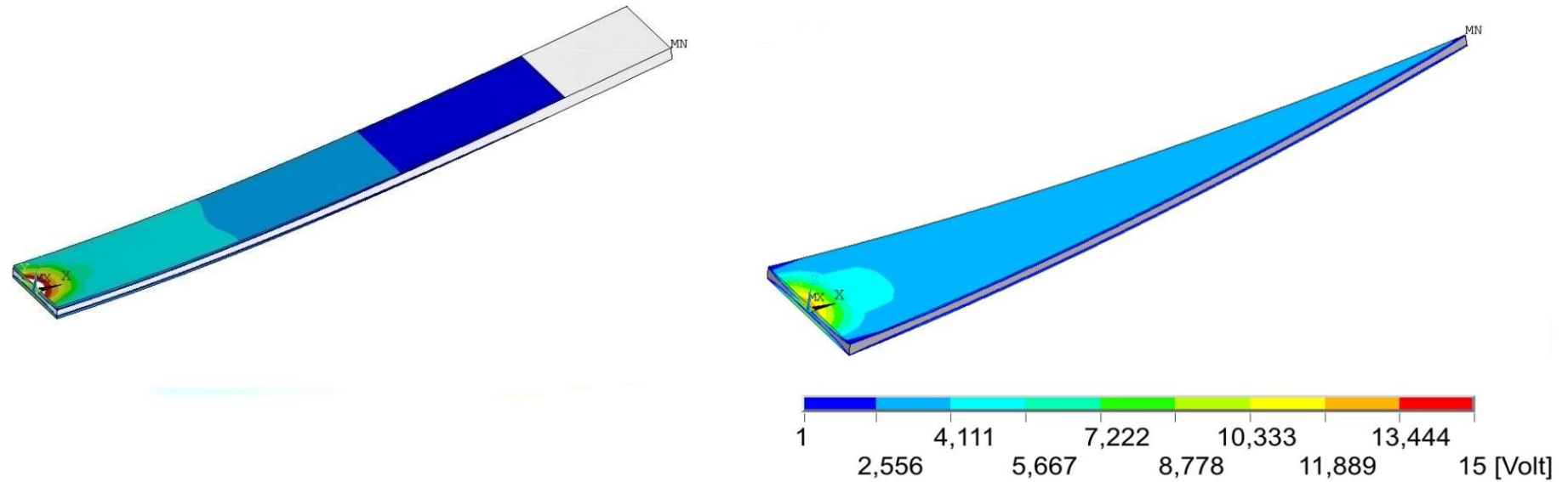
- structured electrodes (complex)
- **stress homogenization**

Kim, Ph.D. Thesis, Univ.
Pittsburgh, USA, 2002



Piezoelectric Energy Harvesters – Examples V

Electrical surface potential at a piezo bimorph



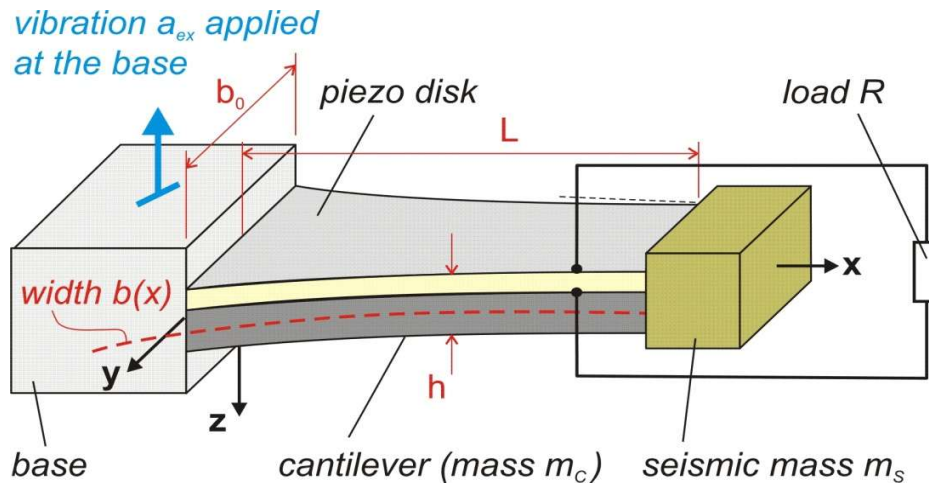
*FEM simulation for a **non-optimized** rectangular shape...*

*... and for an **optimized** triangular shape*

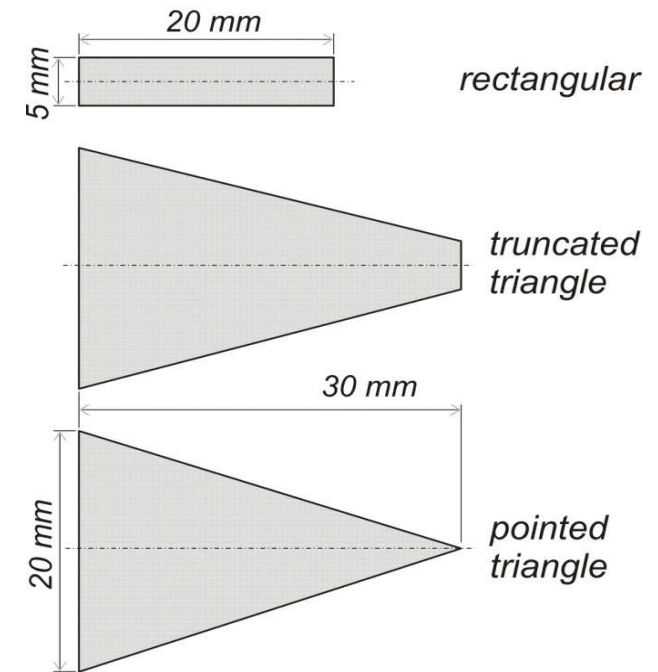
E. Just et al., Proc. GMM-Workshop “Energieautarke Mikrosysteme”, 2006

Piezoelectric Energy Harvesters – Examples VI

Comparison of different beam geometries



Geometries under investigation



Common characteristics

- piezo material:
- piezo thickness:
- substrate material:
- substrate thickness:
- cantilever thickness variation:

Stelco PPK 11

150...200 μm

Stycast 2057 (epoxy polymer)

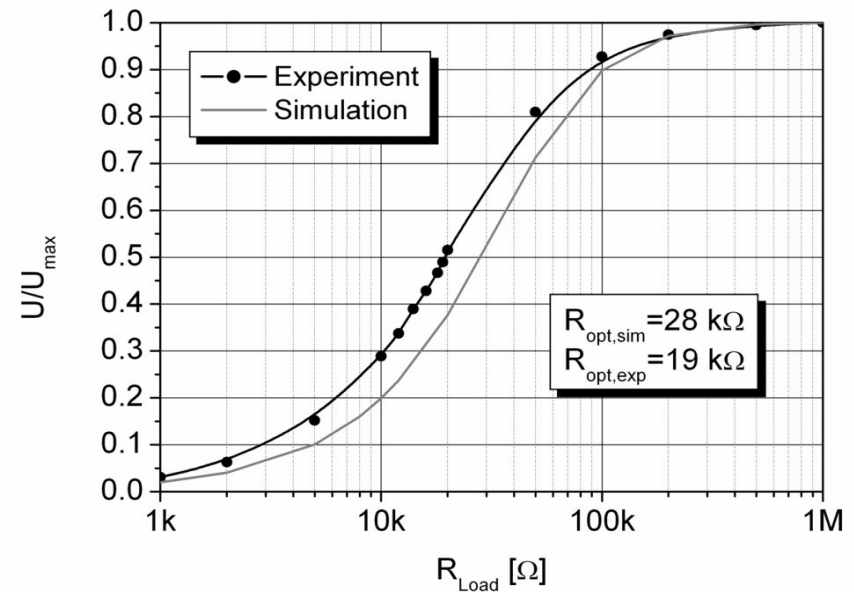
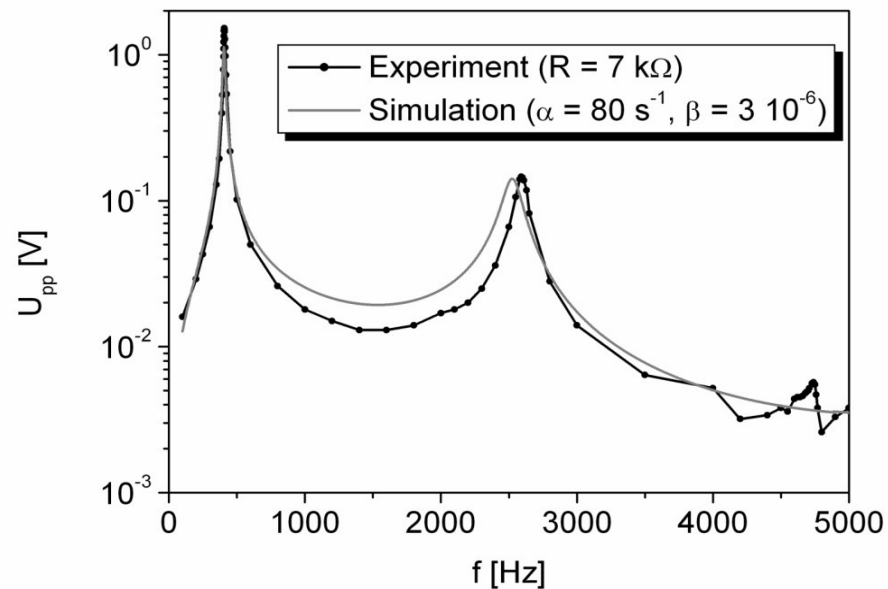
300...400 μm

typ. 20 μm

*F. Goldschmidtboing, P. Woias
J. Micromech. Microeng. 18, 2008,
104013*

Piezoelectric Energy Harvesters – Examples VII

Rectangular beam: Simulation vs. experiment



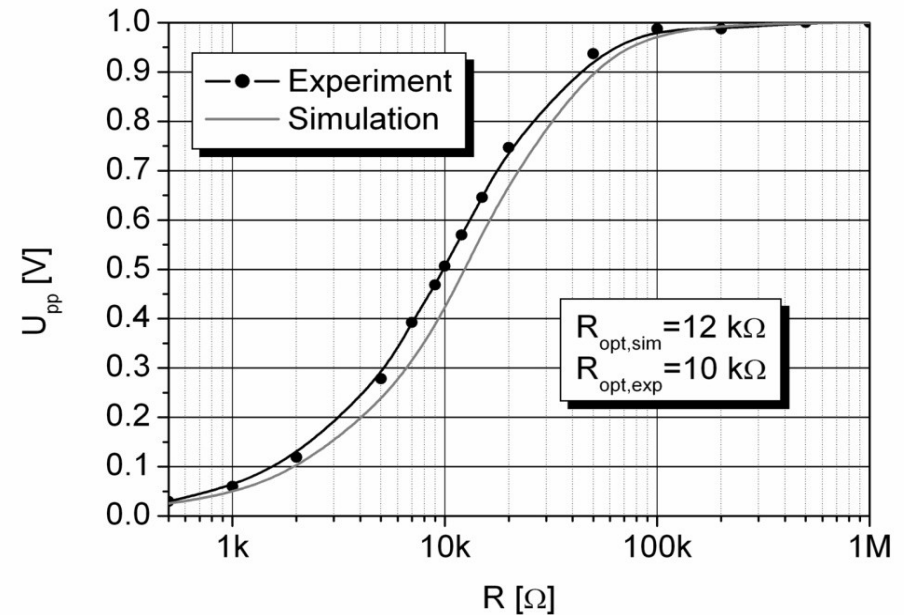
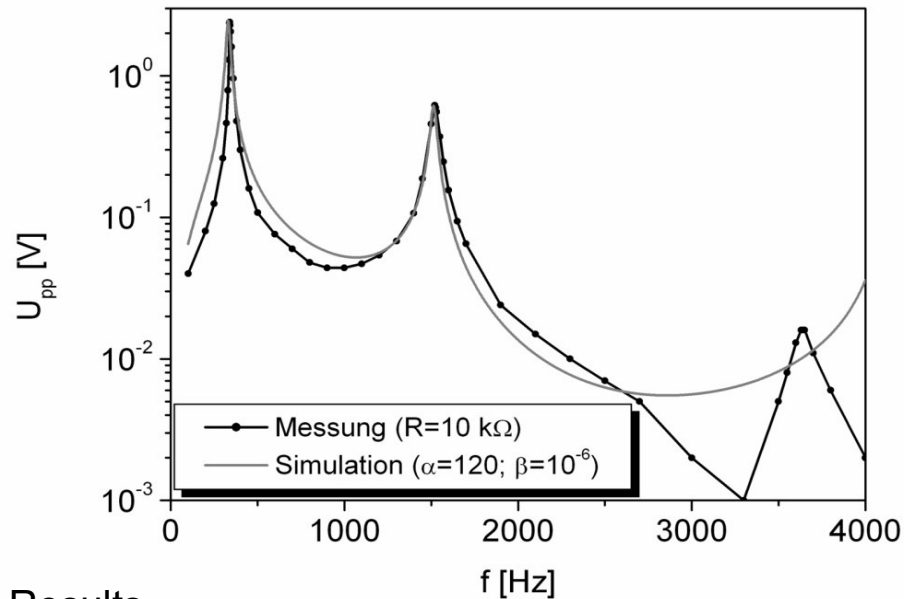
Results

- good agreement for the spectral mechanical response (two 1-D modes resolved)
- only fair agreement for the optimal load resistance

F. Goldschmidtboing, P. Woias, J. Micromech. Microeng. 18, 2008, 104013

Piezoelectric Energy Harvesters – Examples VIII

Triangular beam: simulation vs. experiment



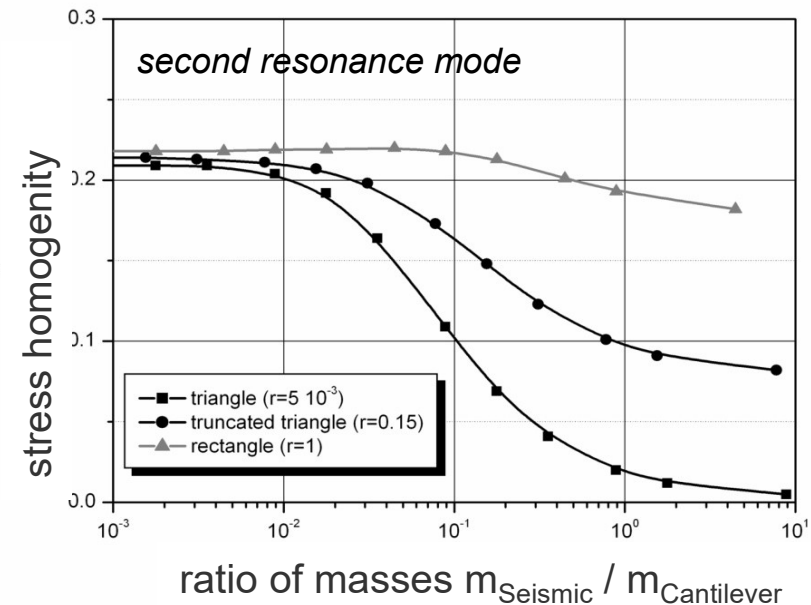
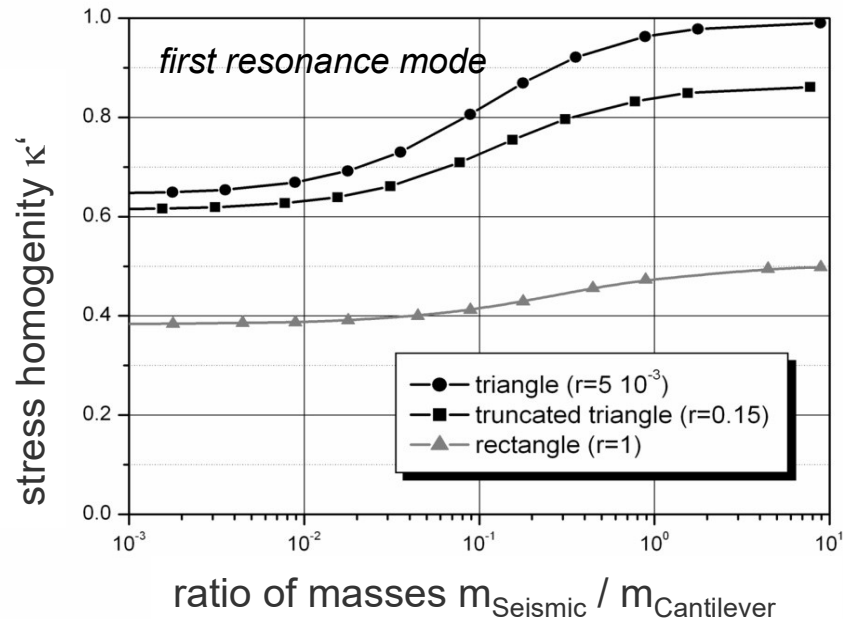
Results

- good agreement for the spectral mechanical response (3rd mode is 2-D)
- good agreement for the optimal load resistance

F. Goldschmidtboing, P. Woias, J. Micromech. Microeng. 18, 2008, 104013

Piezoelectric Energy Harvesters – Examples IX

Influence of the mass ratio ε

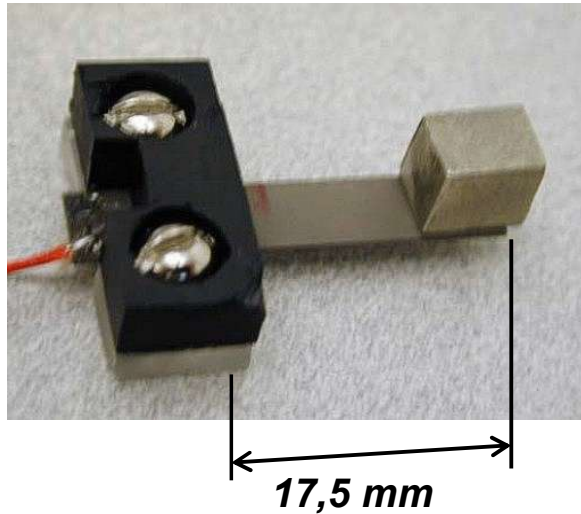


Results

- triangular beam with seismic mass: optimal for 1st modes
- triangular beam w/o seismic mass: better for 1st **and** 2nd modes ?
- **in general:** Optimization has to be done for a certain resonance mode.

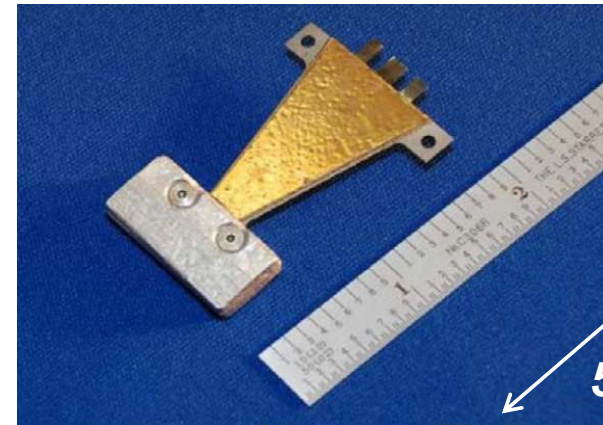
*F. Goldschmidtböing, P. Woias
J. Micromech. Microeng. 18,
2008, 104013*

Piezoelectric Energy Harvesters – Examples



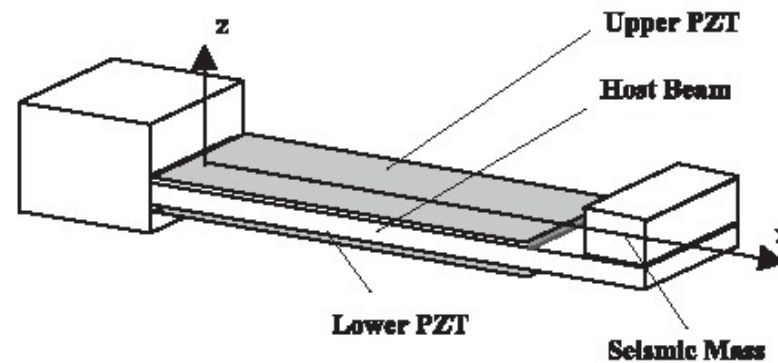
$P_{max} = 0,08 \text{ mW} @ 0,23 \text{ g and } 120 \text{ Hz}$

*S. Roundy et al.,
UC Berkeley, 2004*



$P = 0,45 \text{ mW} @ 1\text{g and } 60 \text{ Hz}$

*„Joule Thief“
© AdaptivEnergy, 2009*



Electrodynamic Harvester – Principle I

Principle: Electromagnetic induction

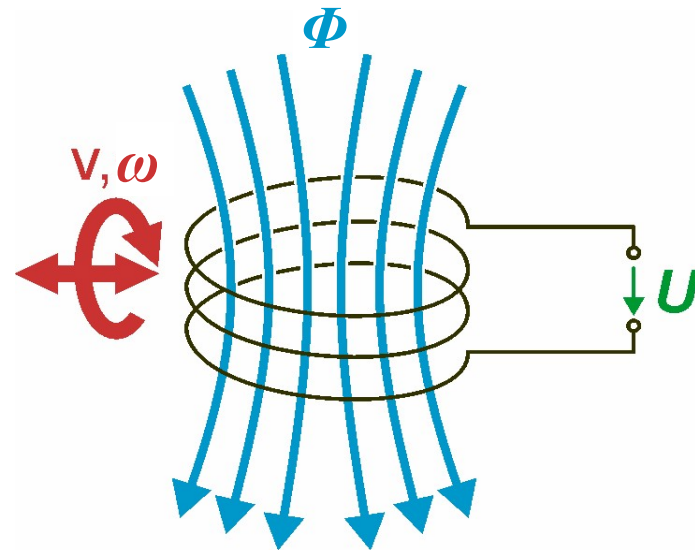
The movement of an electric conductor in a magnetic field with variable flux $d\Phi/dt$ generates an induced voltage at the ends of the conductor.

Properties

- AC currents from motion or induced AC fields
- bad to fair voltage range (mV...V)
- moderate source impedance (<10 kΩ)

Faraday's law of induction

$$U = -N \cdot \frac{d\Phi}{dt}$$

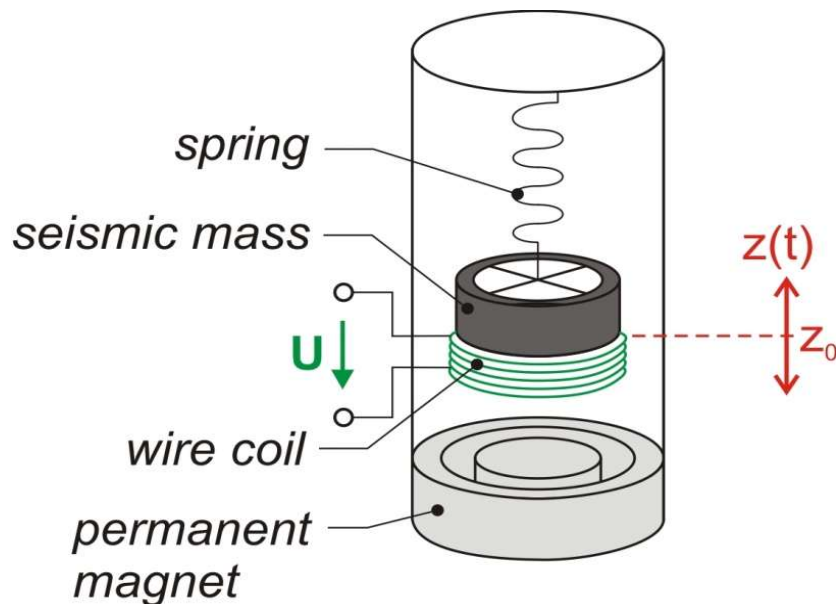


Electrodynamic Harvesters – Principle II

Electrodynamic harvesters: Basic design principles

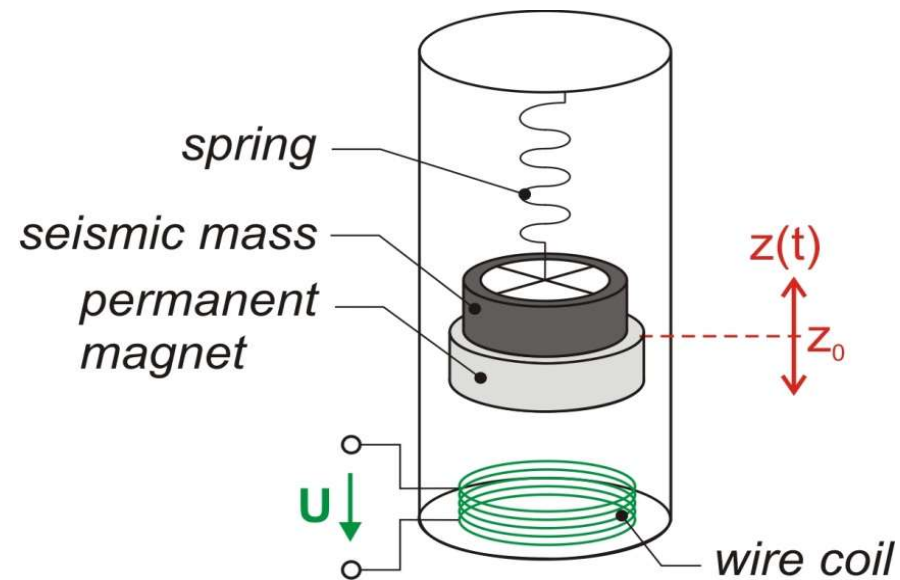
Moving coil

- seismic mass tailored individually 😊
- moving electrical connections ☹️
- possibly a higher total mass ☹️



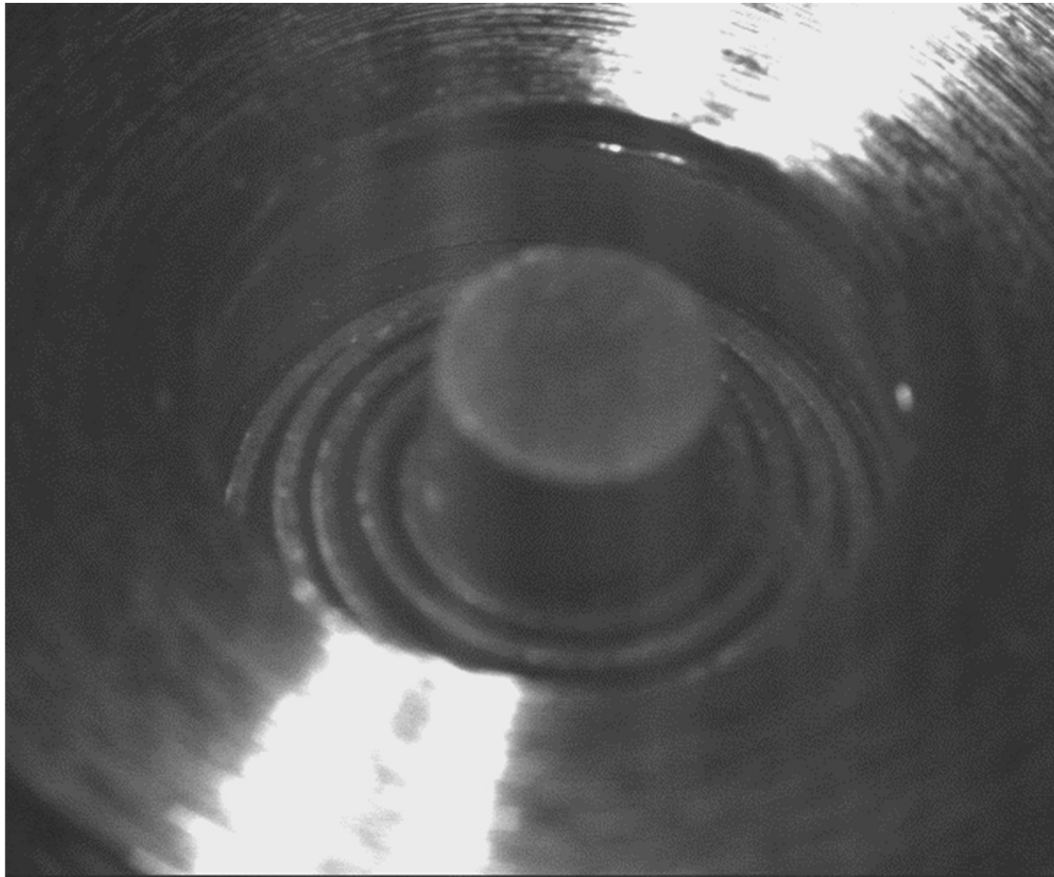
Moving magnet

- Seismic mass = magnet + ... 😊 ☹️
- fixed electrical connections 😊
- possibly a smaller total mass 😊



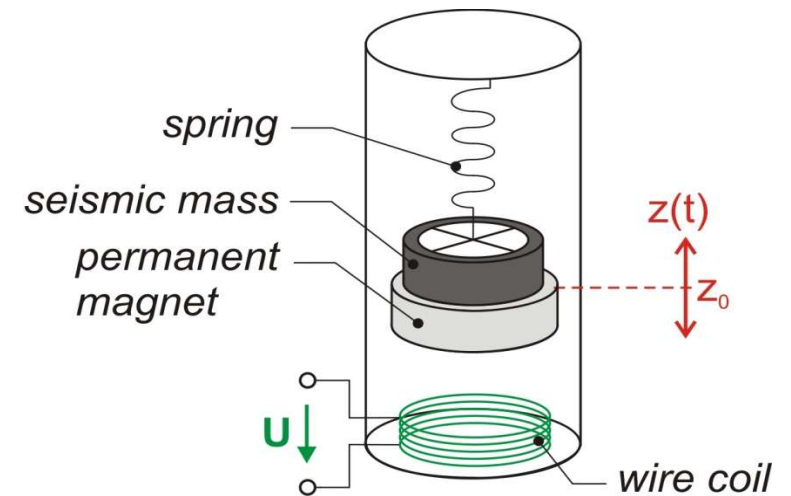
Electrodynamic Harvester – Principle III

Electrodynamic harvesters: Basic design principles



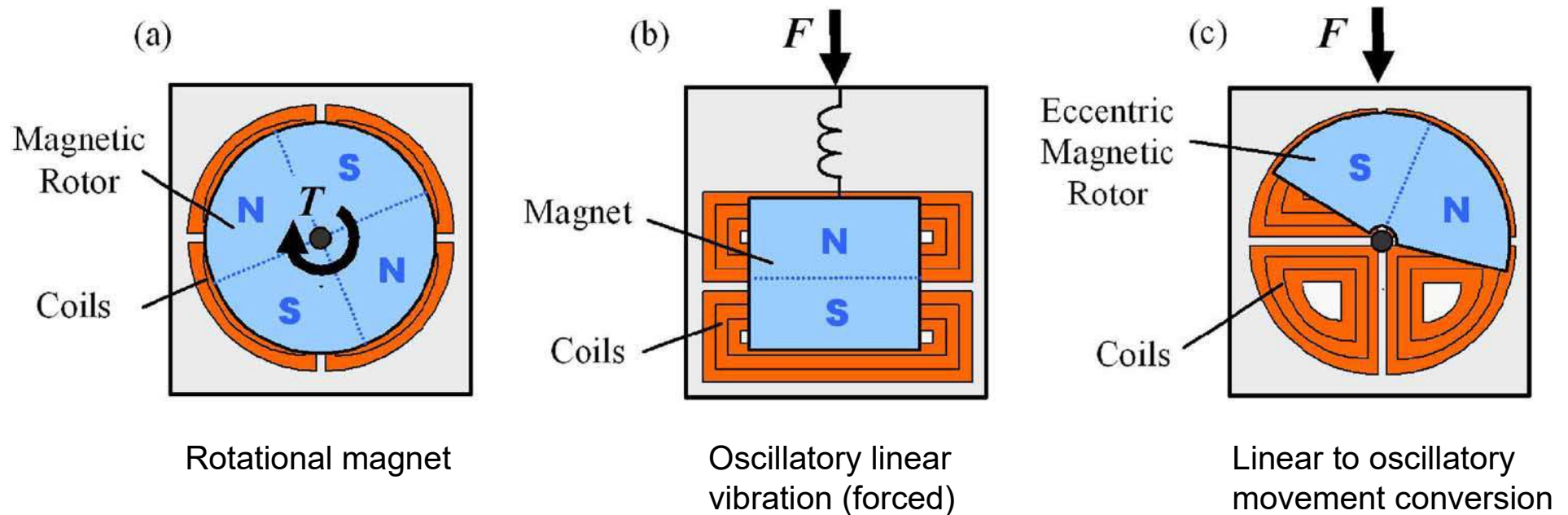
Coil (outside)

Moving magnet (=seismic mass) with spiral spring



Electrodynamic Harvester – Principle III

Different concepts for relative movement possible



D.P. Arnold, Review of Microscale Magnetic Power Generation, Transactions on Magnetics, vol. 43, no. 11, pp. 3940–3951, November 2007.

Electrodynamic Harvester – Modelling I

State variables and material parameters

$\vec{H}$: magnetic field strength $\left[\frac{A}{m} \right]$

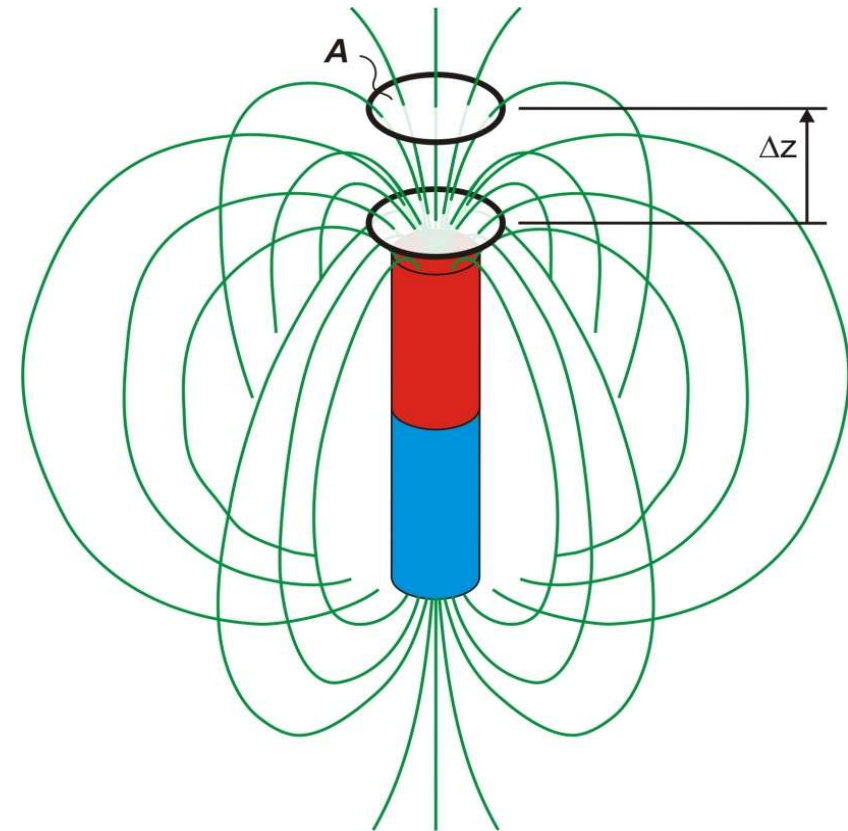
$\vec{B}$: magnetic flux density $\left[\frac{Vs}{m^2} \right]$

μ : magnetic permeability $\left[\frac{Vs}{Am} \right]$

Φ : magnetic flux $[Vs]$

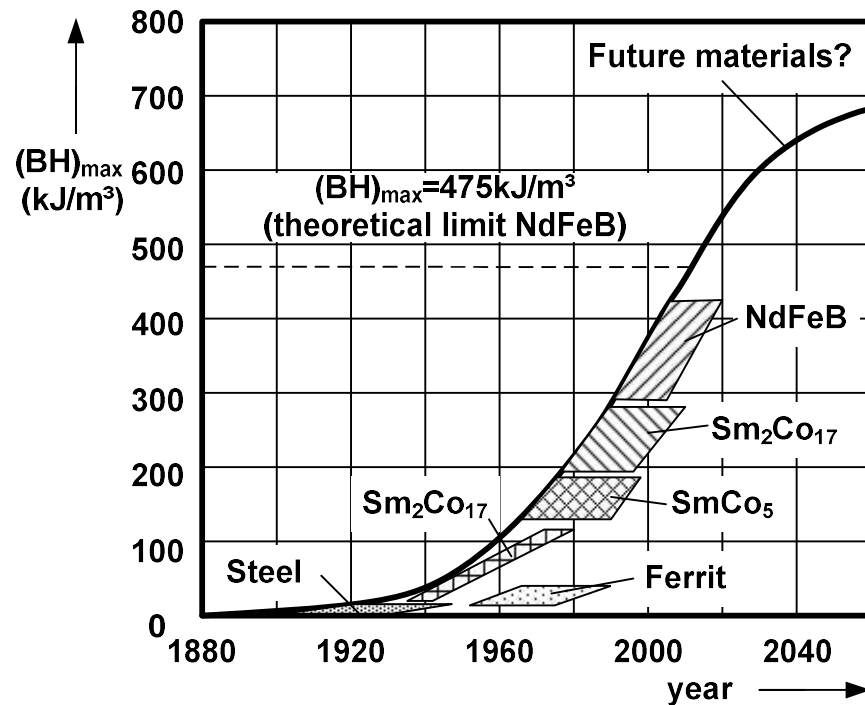
$$B = \mu \cdot H$$

$$\Phi = \int_A \vec{B} \cdot d\vec{A}$$



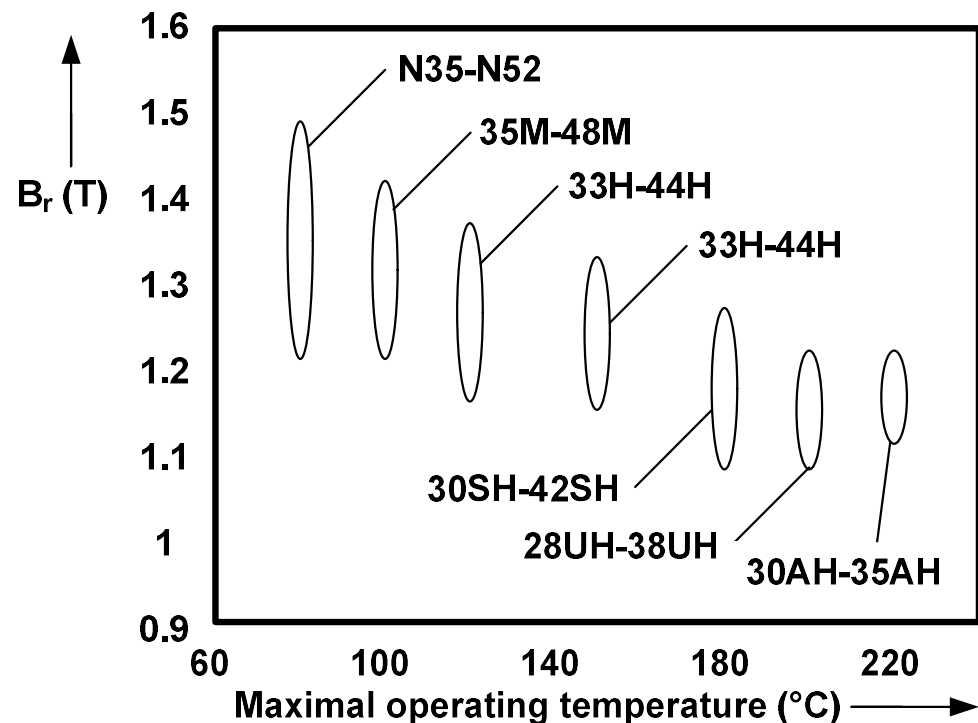
Electrodynamic Harvester – Modelling II

Magnetic materials available:



- Neodymium / FeB best material (up to 1,4 T)
- Neodymium (Nd), Samarium (Sm) rare earth elements

Maximum operating temperatures:



- Occurrence 91 % in China (Nd)
(SPON, „Das neue Gold“, vom 10.04.2009)

Electrodynamic Harvester – Modelling III

Electromagnetic transduction factor

What is the electromagnetic *transduction factor* k ?

- relationship between induced voltage U and movement of seismic mass (or coil)
- Relationship between induced force on the seismic mass to current through the coil

$$k = \frac{U}{\dot{z}} = \frac{F}{I}$$

Faraday's law:
$$U = -N \cdot \frac{d\Phi}{dt} = -N \cdot \frac{d\Phi}{dz} \cdot \frac{dz}{dt} = -k(z) \cdot \dot{z}(t)$$

Assumption 1: $B \sim$ constant over A

$$\Phi = \int_A \vec{B} \cdot d\vec{A} \approx B(z) \cdot A \Rightarrow U = -N \cdot \underbrace{\frac{dB}{dz}}_{k(z)} \cdot A \cdot \dot{z}(t)$$

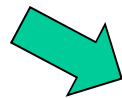
$$k = N \cdot A \cdot \left. \frac{dB}{dz} \right|_{z_0} \approx \text{const.}$$

Assumption 2: small movements dz around z_0 :

How to match the load?

- Electrical energy is partly consumed in the load
- and partly wasted in the coil

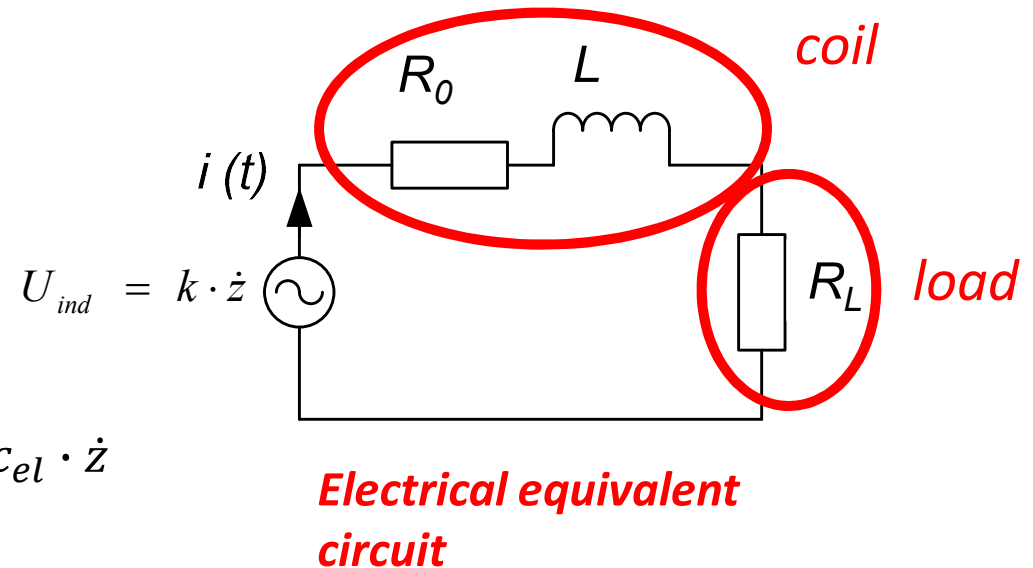
$$F_{damp,el} = k \cdot I = k \cdot \frac{U}{R_0 + R_L} = \frac{k^2}{R_0 + R_L} \cdot \dot{z} = c_{el} \cdot \dot{z}$$



$$P_{el}(c_{el}) = \frac{1}{2} \cdot \frac{c_{el}}{(c_{el} + c_{mech})^2} \cdot m^2 \cdot a_{vib}^2 \cdot \frac{R_L}{R_0 + R_L}$$



$$R_{L,opt} = R_0 + \frac{k^2}{c_{mech}}$$



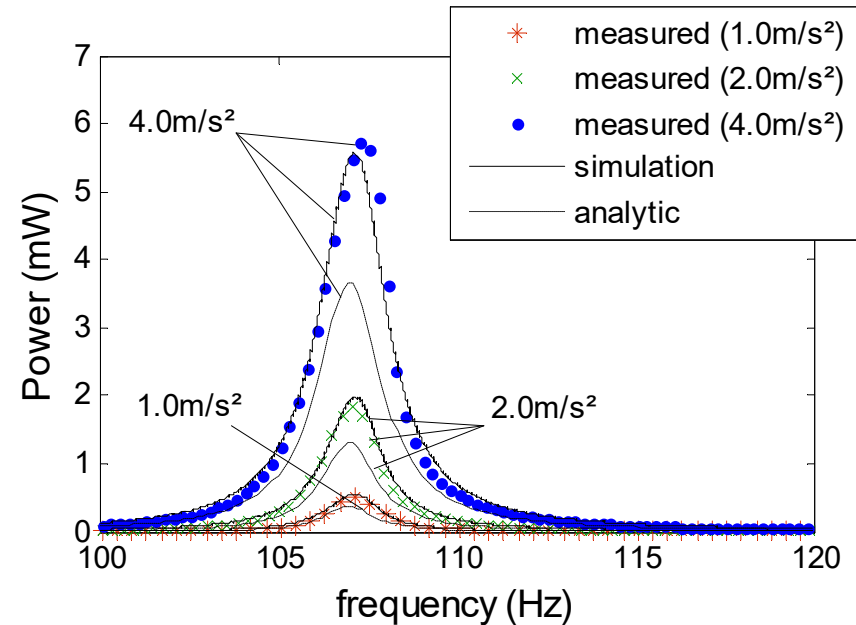
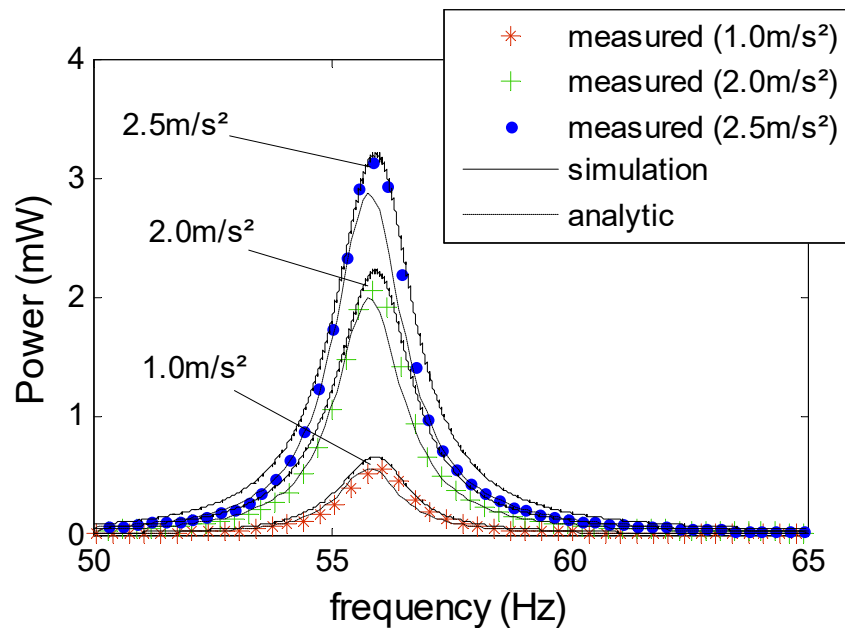
Electrodynamic Harvester - Results I

Measurement results and application scenario

Properties for two optimized generators:

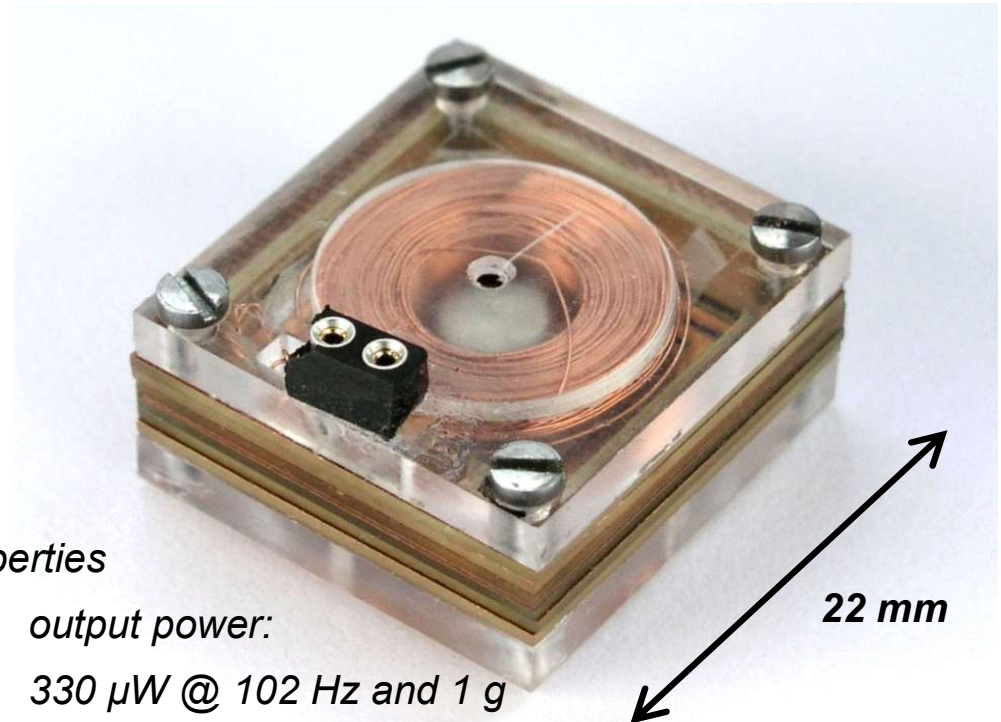
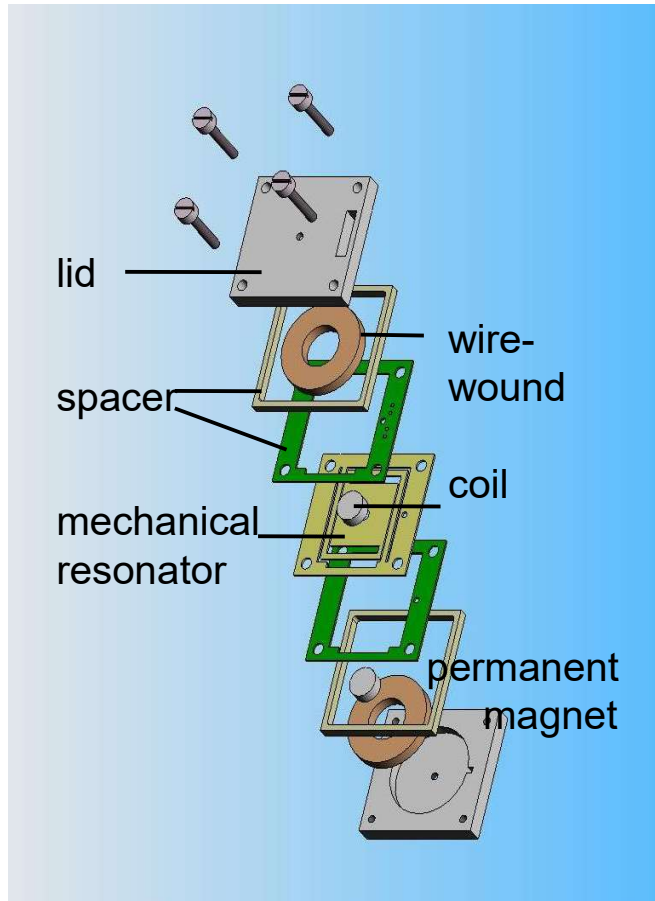
- Up to 3 mW / 5.5 mW at resonance

D. Spreemann et al., IMTEK -
Lehrstuhl Mikroelektronik (Diss)



Electrodynamic Harvester - Results II

Small electrodynamic harvesters



Properties

- output power:
- $330 \mu\text{W}$ @ 102 Hz and 1 g
- no-load voltage: 210 mV
(improved via modified coil design)

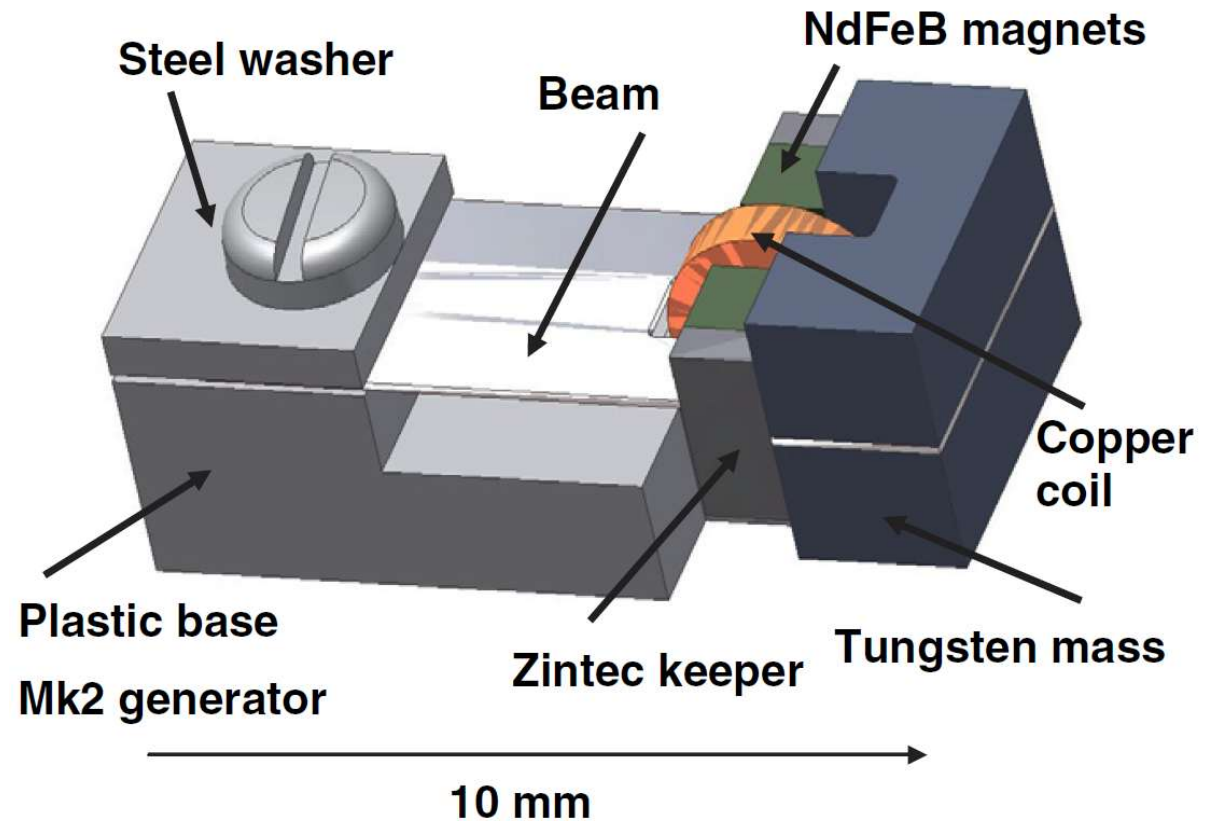
E. Bouendeu, J. Korvink, IMTEK – Laboratory for Simulation

Electrodynamic Harvester - Results III

Small electrodynamic harvesters

Mk2-generator design:

- Magnet size: $2.5 \times 2 \times 1.5 \text{ mm}^3$
- Coil turns: 2300
- Wire diameter: $12 \text{ }\mu\text{m}$
- Coil resistance: $1530 \text{ [}\Omega\text{]}$
- Open-circuit voltage: $595 \text{ mV}_{\text{rms}}$
- Optimum load resistance: $4 \text{ k [}\Omega\text{]}$
- Output Power @ 0.6 m/s^2 , optimum load resistance:
 $45 \text{ }\mu\text{W}_{\text{rms}}$

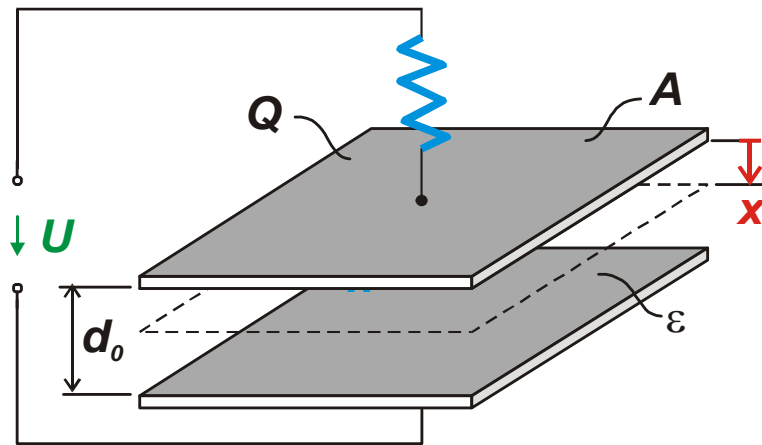


Torah *et al*, Meas. Sci. Technol. **19**, (2008), 125202 R

Electrostatic Harvesters

General Properties

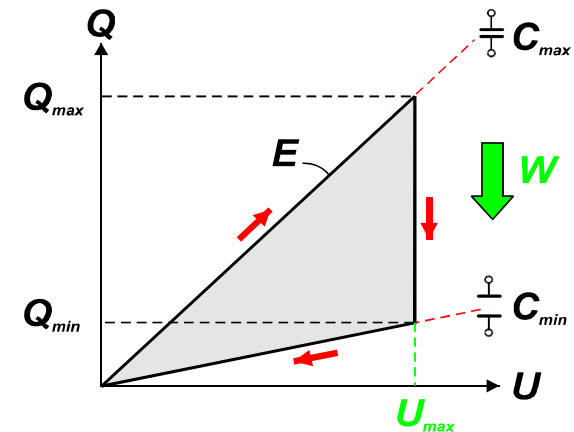
- AC currents from a dynamic change of capacitance
- some need an auxiliary voltage (except electret type)
- fair voltage range (1 V...100 V)
- very high source impedance (> 1 MOhm)



$$\begin{aligned} dW &= U \cdot dQ \\ Q &= C \cdot U \end{aligned}$$

$$C = \epsilon_0 \epsilon_R \cdot \frac{A}{d}$$

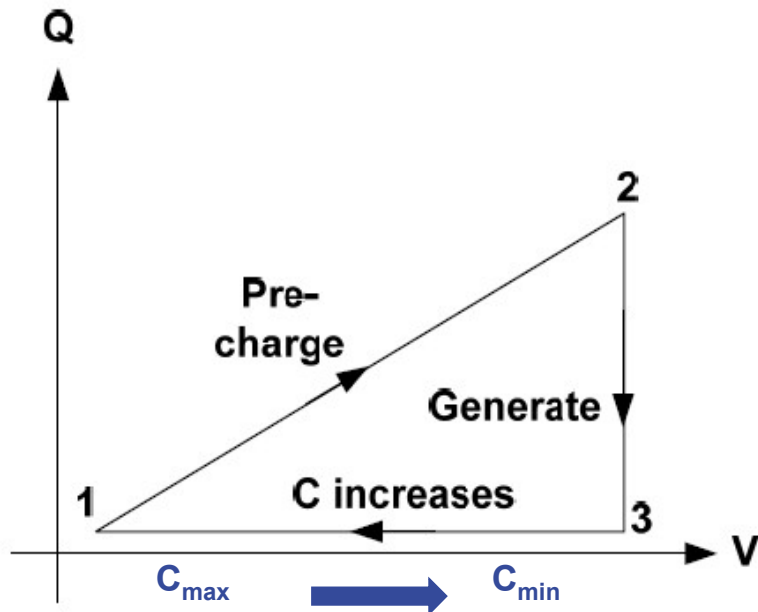
$$\Rightarrow W = \frac{1}{2} \cdot (C_{max} - C_{min}) \cdot U_{max}^2$$



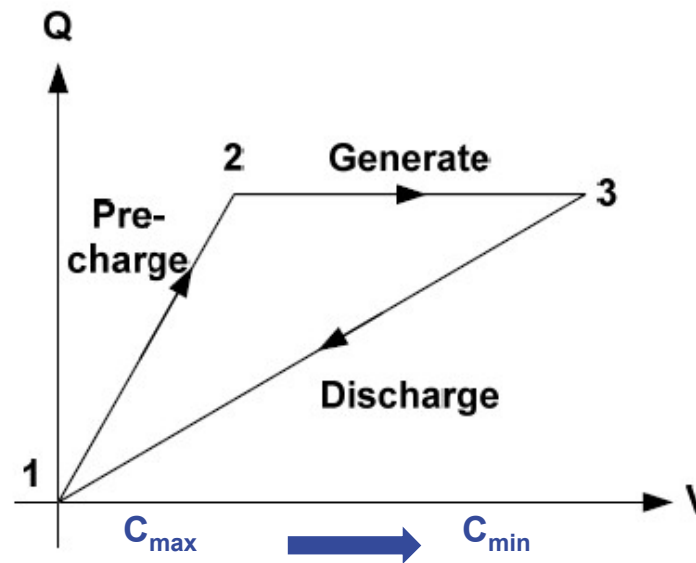
Electrostatic Harvesters

Basic Principles of Operation

Constant **voltage** operation



Constant **charge** operation:



Mitcheson et al, *Electrostatic Microgenerators*, Meas. control-UK, 2008

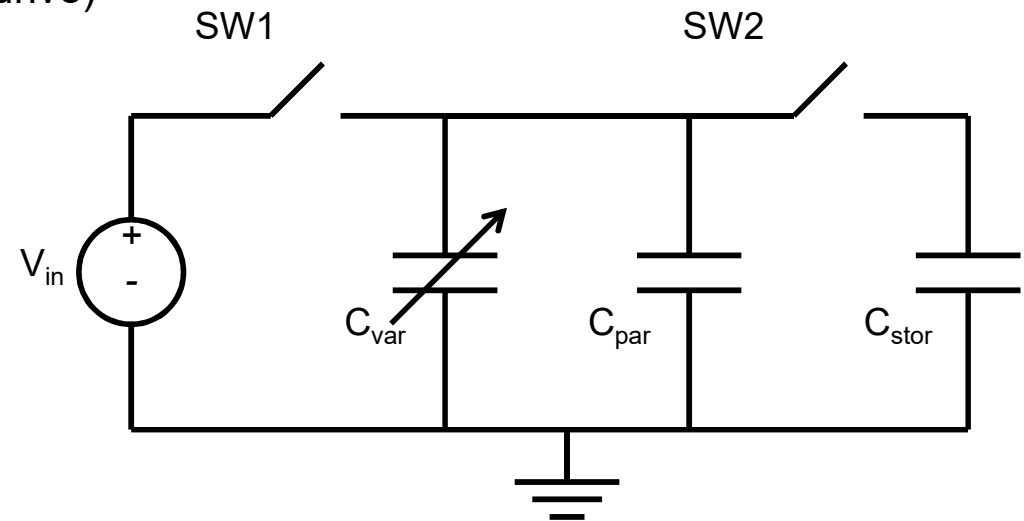
Electric Circuit System Model

Constant Charge Operation

- Pre-charged reservoir (battery or capacitor) represents the voltage source
- C_{var} is the variable capacitance structure (i.e. comb drive)
- C_{par} is the parasitic capacitance
- C_{stor} is the storage capacitor
- SW1 & SW2 are switches (initially open)
- SW1 is closed when $C_{\text{var}} = C_{\text{max}}$ (V_{in} across C_{var})
 - → Charge transfer to C_{var} ($Q = C_{\text{max}} + C_{\text{par}} \cdot V_{\text{in}}$)
- SW1 opens ; C_{var} moves to minimum C_{min}
 - → Energy stored in C_{var} goes up
- @ C_{min} SW2 closes and transfers charge to C_{stor}

$$W = \frac{1}{2} \cdot C \cdot U^2 = \frac{1}{2} \cdot \frac{1}{C} \cdot Q^2$$

Circuit representation for an electrostatic converter



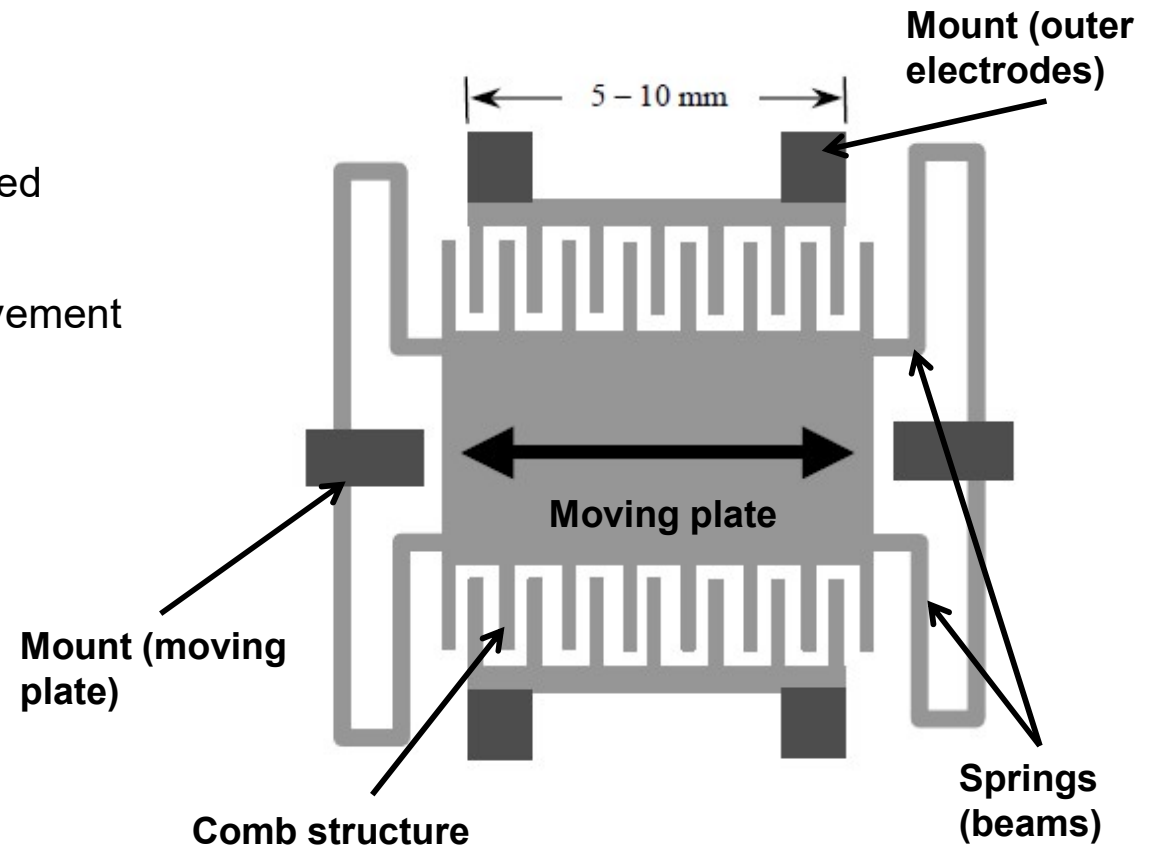
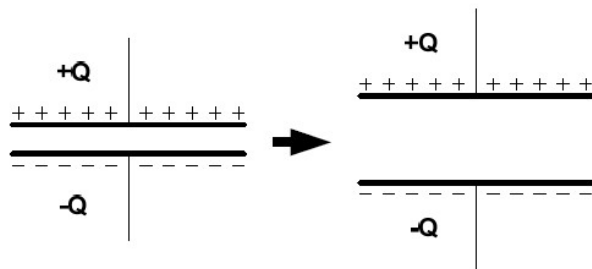
→ *Mechanical energy converted to electrical energy*

Roundy S. J., 2003, PhD thesis

Basic MEMS Design of Electrostatic Harvesters

Horizontal In-plane Design

- Change of capacitance by changing the gap between the electrode fingers
- Moving plate with the comb structures mounted by springs (bending beams)
- Springs designed for optimum horizontal movement
- Spring design:
 - Horizontal parallel movement
 - Linear behavior
 - No torsion

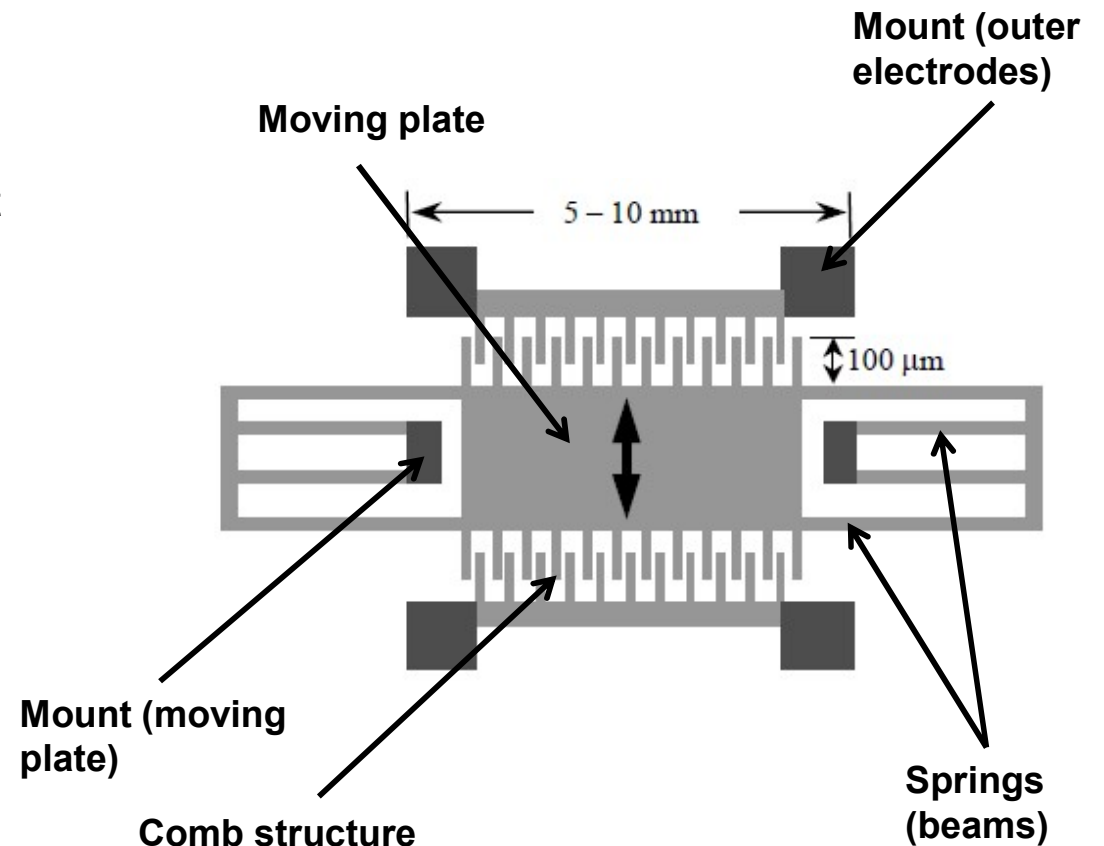
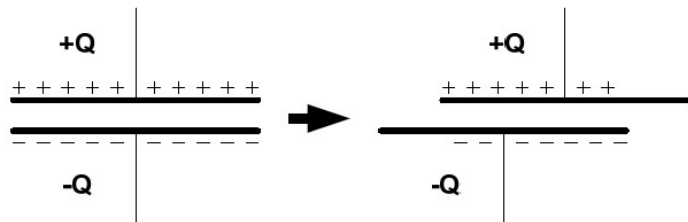


Roundy S. J., 2003, PhD thesis

Basic MEMS Design of Electrostatic Harvesters

Vertical In-plane Design

- Change of capacitance by changing the overlapping area of the comb electrodes
- Springs designed for optimum vertical movement
- Spring design:
 - Parallel movement in vertical direction
 - Linear behavior
 - Torsion must be prohibited (gap between combs typically only some μm)

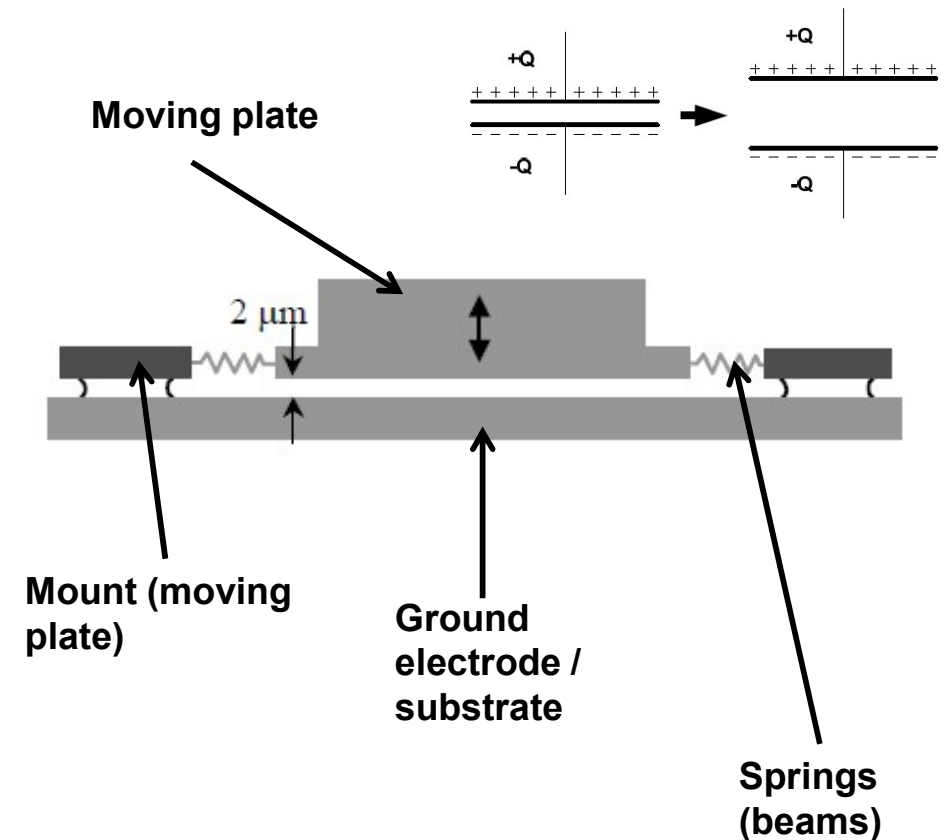


Roundy S. J., 2003, PhD thesis

Basic MEMS Design of Electrostatic Harvesters

Out-of-plane Design

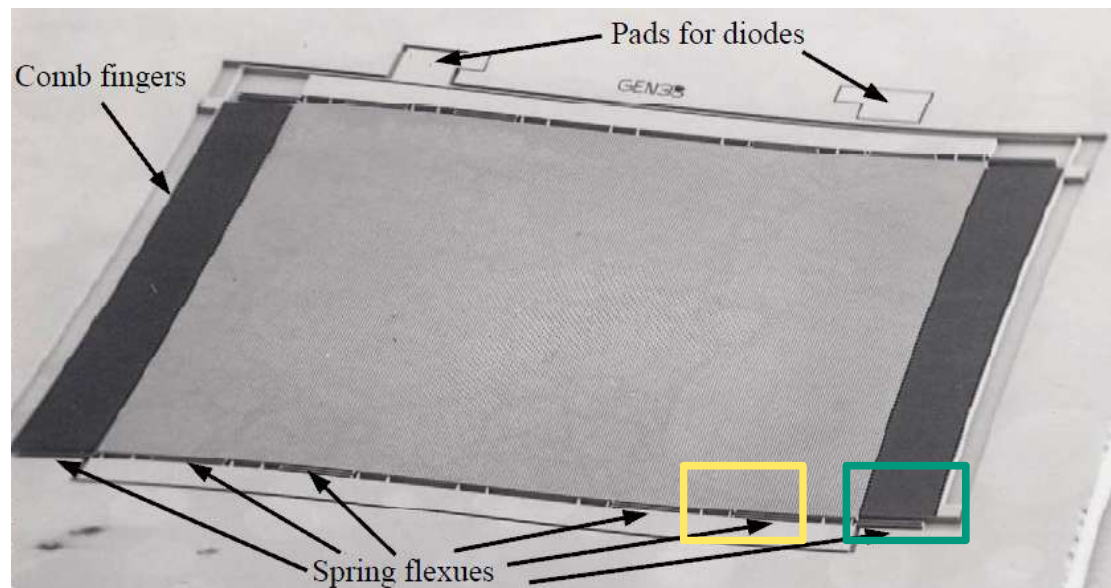
- Change of capacitance by changing the gap between the moving plate and the underlying bottom electrode / substrate
- Springs designed for out-of-plane movement
- Spring design:
 - Parallel movement in vertical direction
 - Linear behavior
 - Torsion unwanted but not as critical due to possible parallel spring design



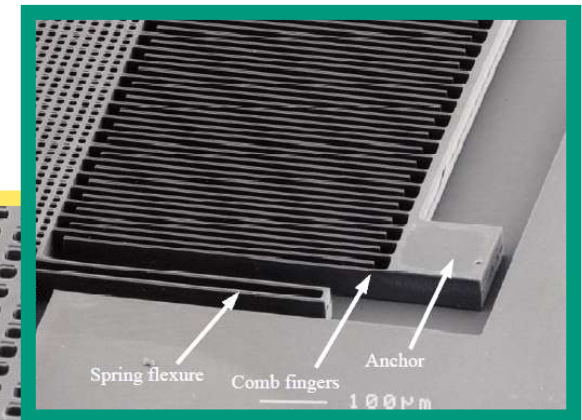
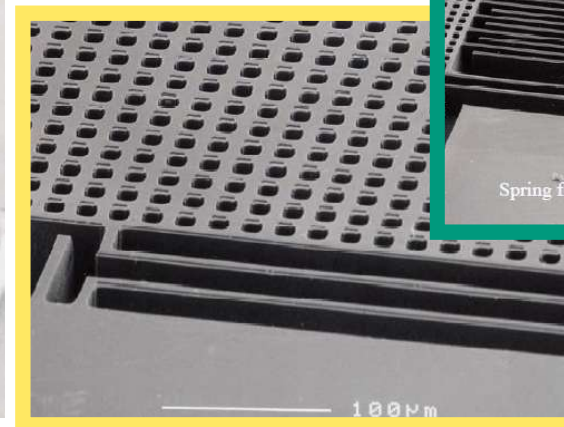
Roundy S. J., 2003, PhD thesis

Design of Electrostatic Harvesters

Electrostatic In-plane Harvester (Roundy)



Electrode fingers & center plate



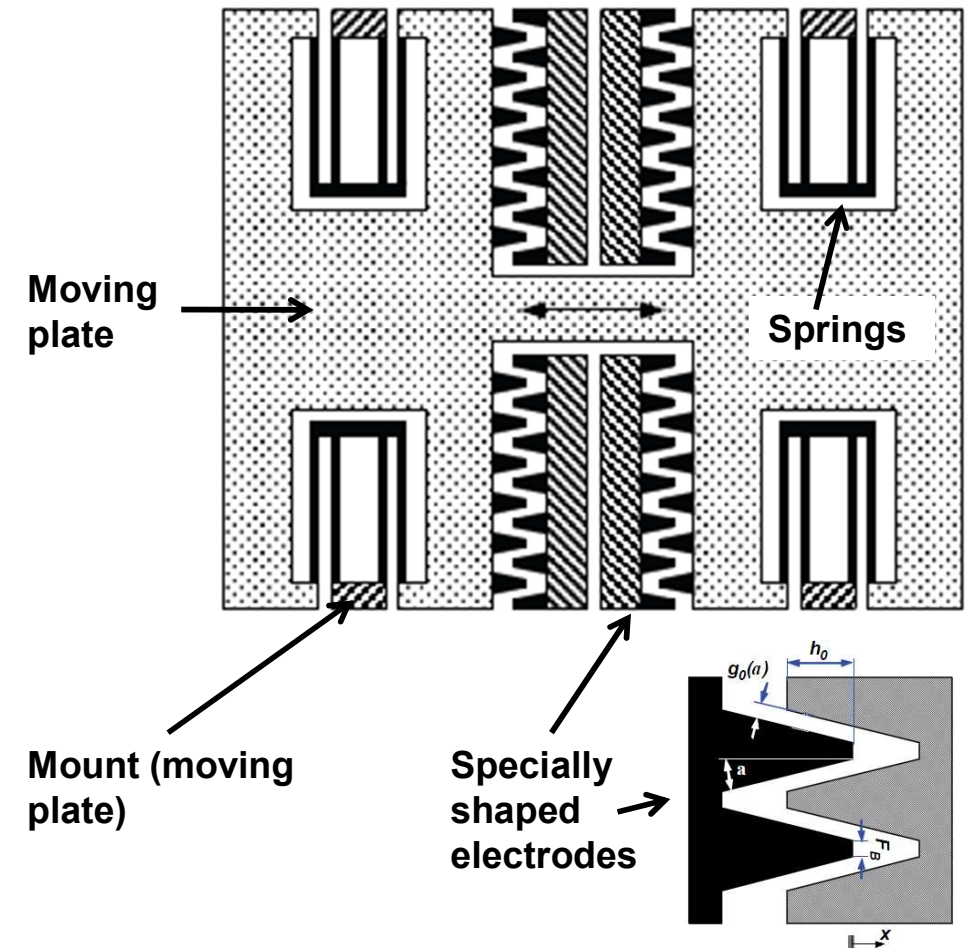
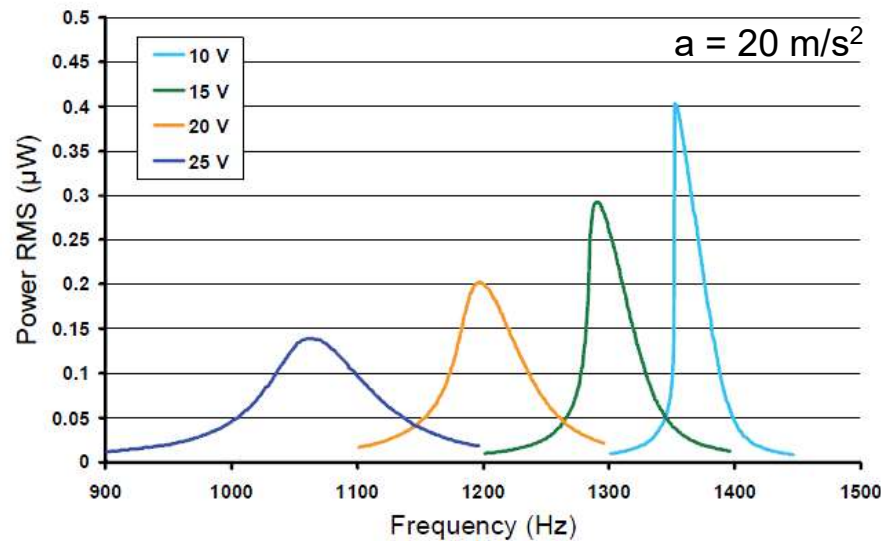
Predicted output power for optimum parameters and a (large) tungsten seismic mass @ 10 V input :

- Up to $\sim 116 \mu\text{W}$ $\rightarrow$ Operation frequency ?

Design of Electrostatic Harvesters

In-plane Gap-closing Converter

- Advanced design with optimized electrode shape
- Bias voltage 10...25 V
- Output power 0.1...0.8 μW
- Tunable by bias voltage in a large frequency range

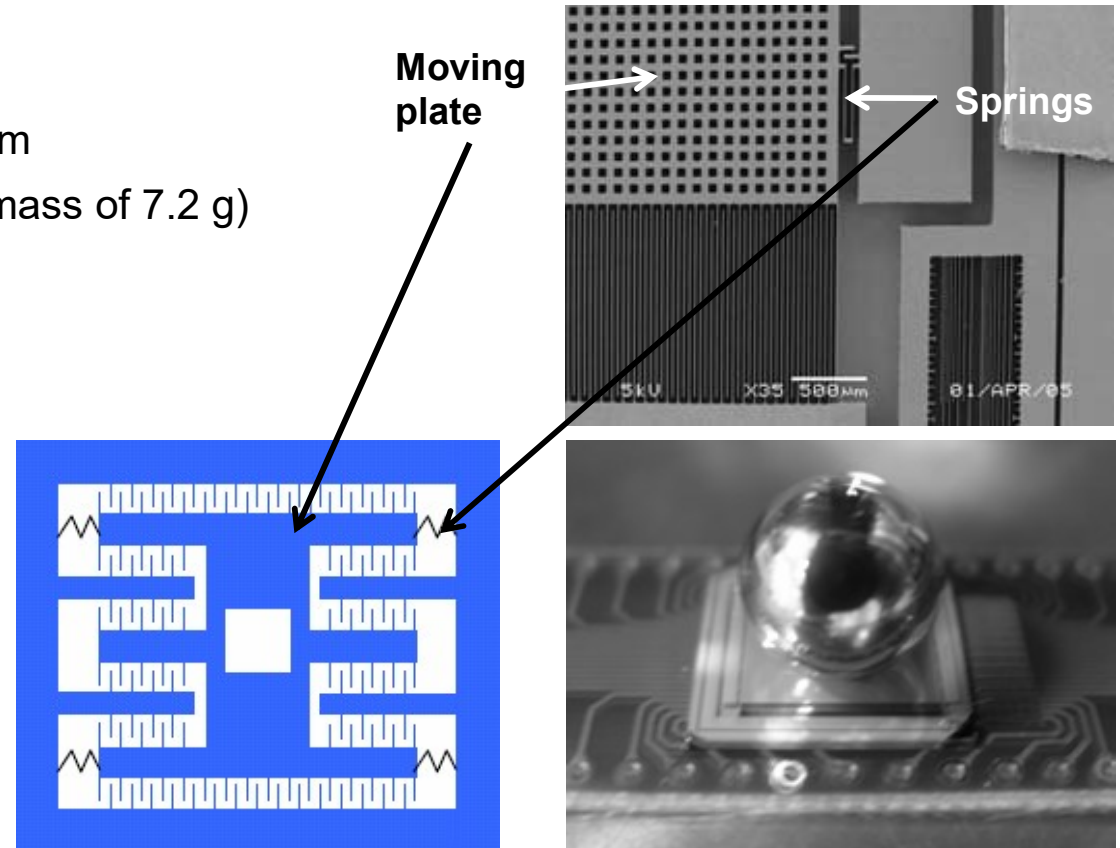
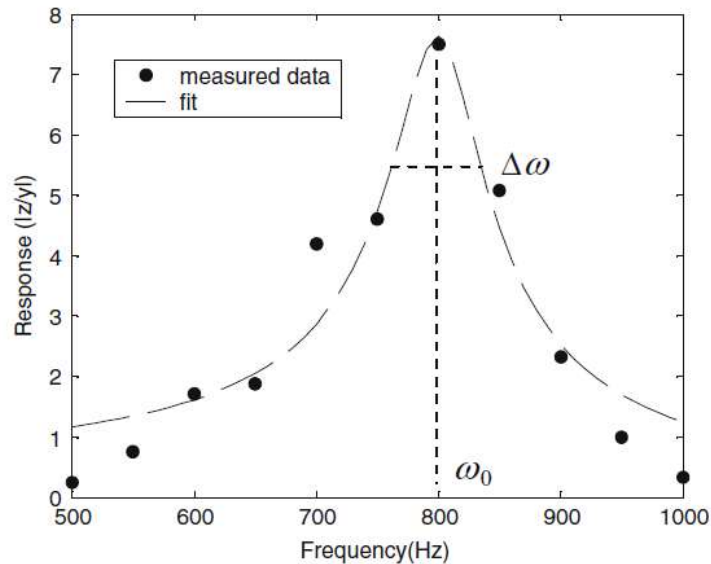


Hoffmann, 2010, IMTEK – LS Manoli / HSG IMIT

Design of Electrostatic Harvesters

In-plane Gap-closing Converter

- Mechanical stop structures
- SOI process, electrode length 200 μm , width 6.8 μm
- Predicted output power of 200 μW (with attached mass of 7.2 g)

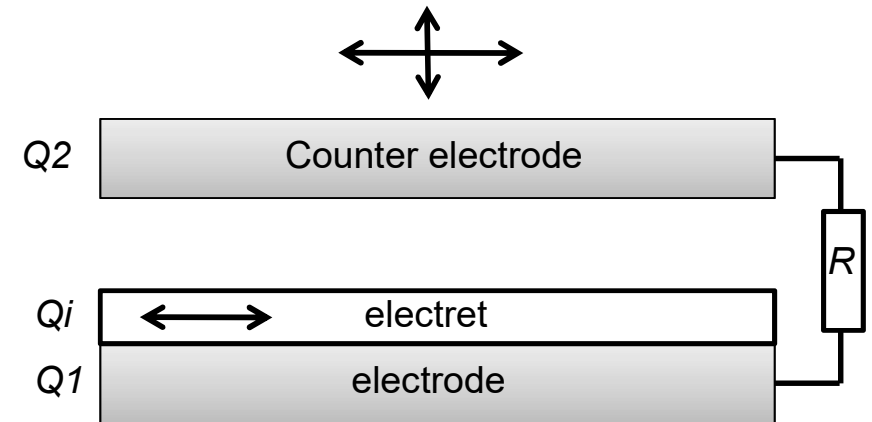


Chiu et al, DTIP 2006, National Chiao Tung University

Electret-based Electrostatic Harvesters

Basic Principle

- Instead of charging a capacitor a pre-charged (or primed) electret can be used
- Charge is transferred during fabrication of the harvester to the electret
- Same principle as in flash memory
- Charge is predicted to stay inside the electret up to 50 years
- Up to 300 V can be generated in the electret
- Advantage: no need for a charge or voltage source

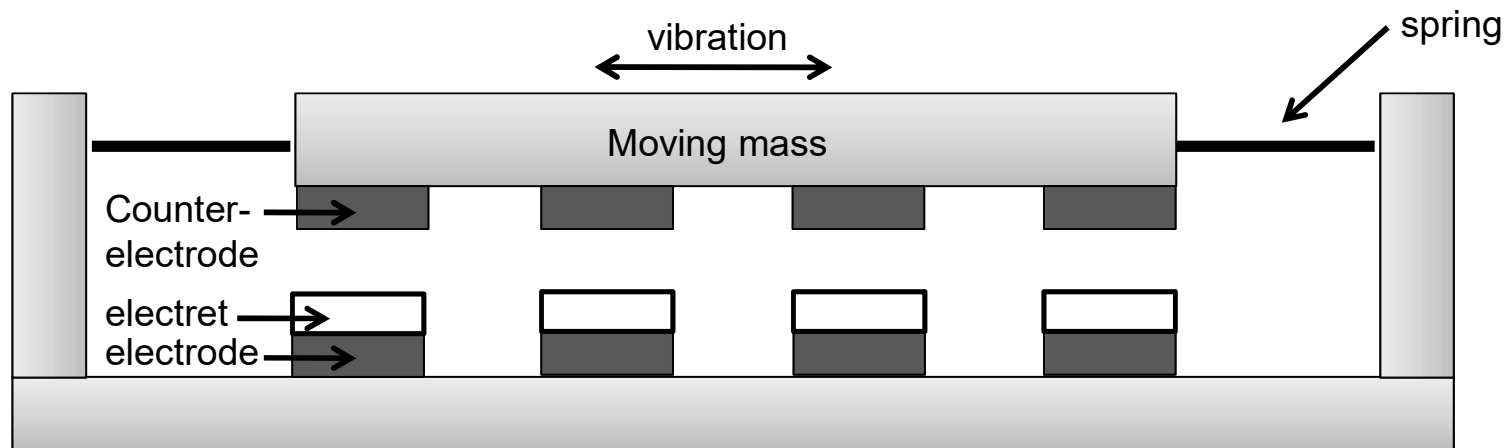


- Parallel or horizontal movement of the electrodes
- Change of capacitance according to the movement generates current over R
- Instead of electrodes, also the electret can be moved

Basic MEMS Design of Electret Harvesters

Overview

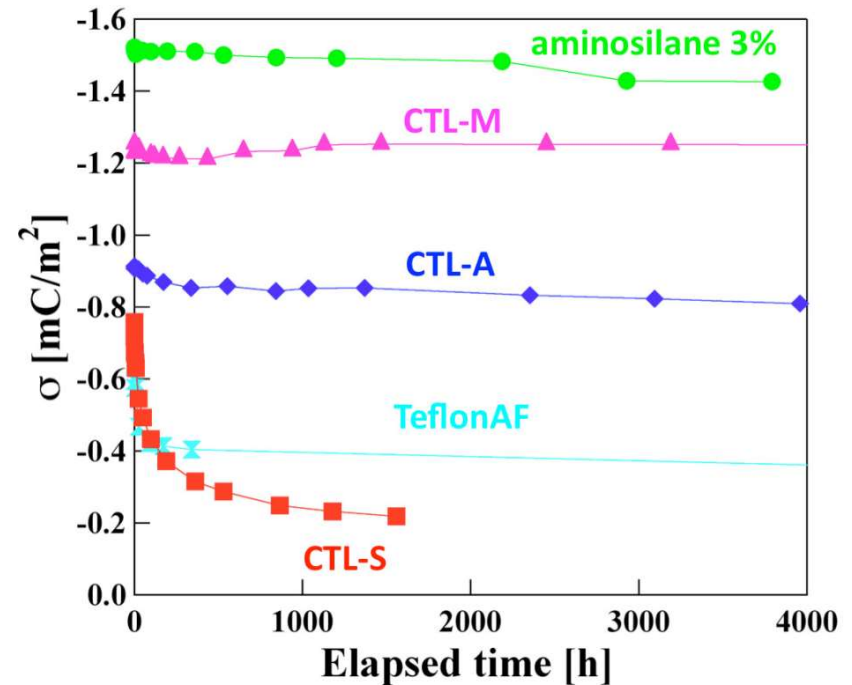
- Moving mass with counter electrodes attached to the frame (substrate) by springs
- Patterning of electrodes for optimization of electrostatic force and output power
- If electrostatic force is too high movement of the seismic mass is inhibited



Electret Harvesters

Electret Materials

- SiO₂ based
 - ✓ High surface charge densities
 - ✓ Possibly bad long-term stability
 - ✓ Multiple layers needed (+Si₃N₄) for good insulation
- Polymer-based:
 - Parylene
 - Teflon
 - CYTOP
 - ✓ Good long-term stability
 - ✓ Partially not MEMS compatible
 - ✓ Difficult to integrate into fabrication process



Surface charge density of different polymers measured over time

Suzuki, Proc. PowerMEMS 2010, The University of Tokio

Electret Harvesters

Material Properties

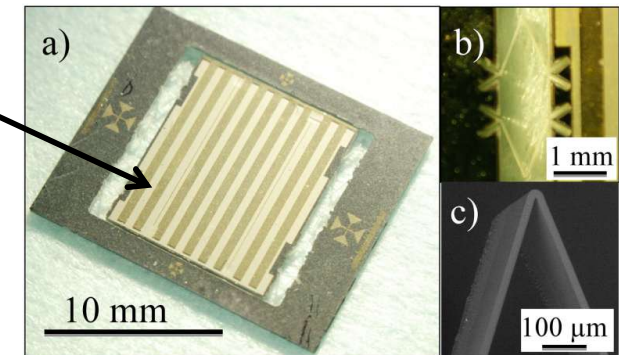
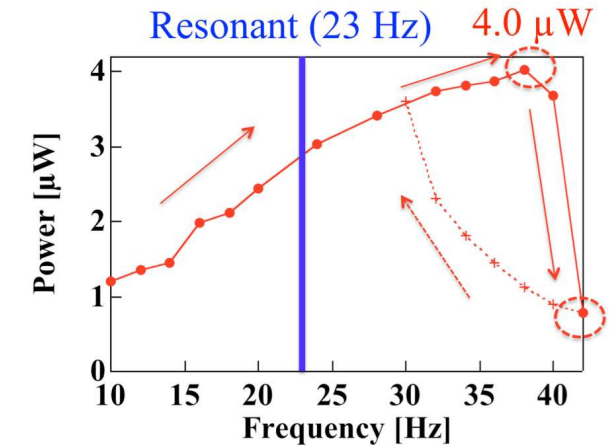
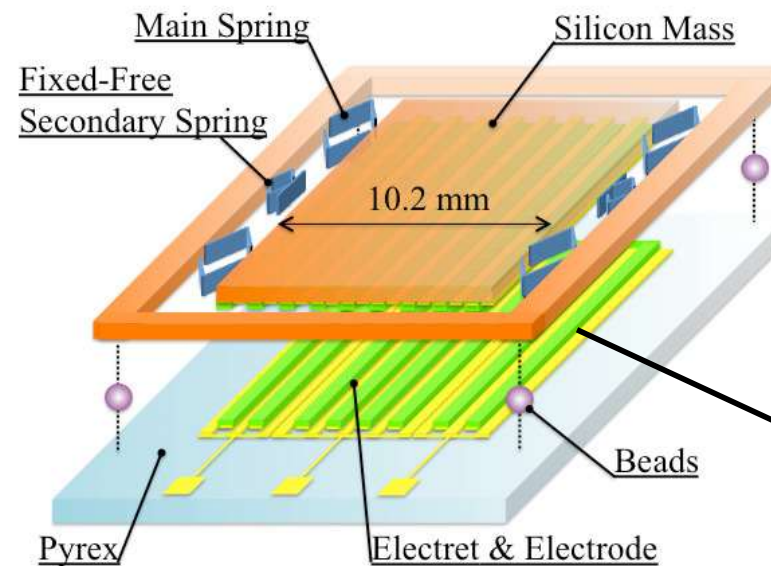
Properties of different electret materials used in energy harvesters:

	CYTOP	Teflon AF	PTFE	Thermal SiO ₂	LPCVD Si ₃ N ₄
Dielectric constant	2.1	1.9	2.1	3.9	6-7
Volume resistivity (Ohm cm)	10 ¹⁷	>10 ¹⁹	>10 ¹⁸	10 ¹⁶	10 ¹⁴
Dielectric strength (kV mm ⁻¹)	110	21	18	500	1000
Max. charge density (mC m ⁻²)	0.68	28		16	

Mescheder et al, Properties of SiO₂ electret films charged by ion implantation for MEMS-based energy harvesting systems, JMM 19 (2009)

Design of Electret-based Micro Energy Harvesters

- Seismic mass: Si, 0.4 g
- Resonance frequency 23 Hz
- Input amplitude 10 m/s²
- Power output at resonance ~ 4 μ W
- Wafer separation by micro beads

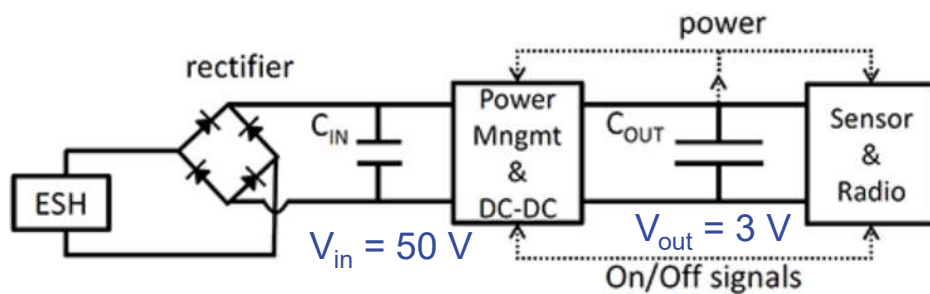


Suzuki, Proc. PowerMEMS 2010

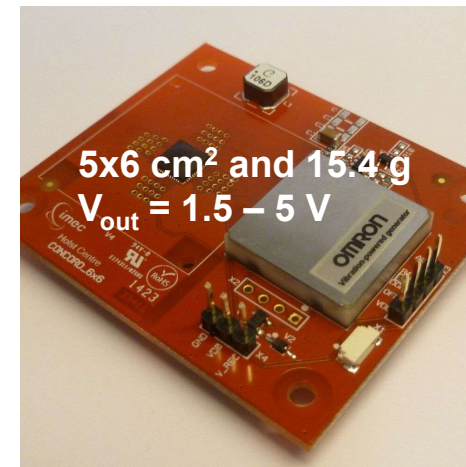
Design of Electret-based Micro Energy Harvesters

First Commercial Harvester

- Developed by OMRON & imec
- PMMC for managing tire pressure from imec:



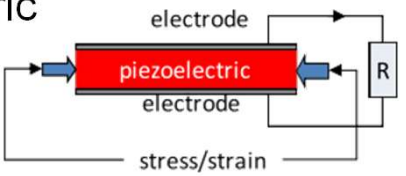
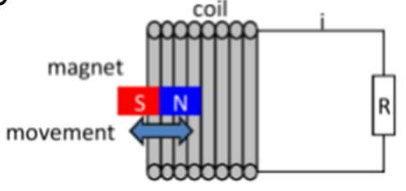
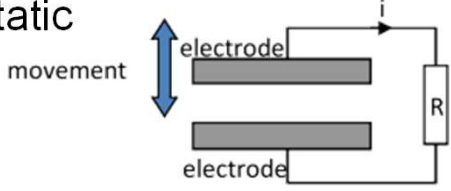
- - Sub-threshold design (CPUs from Ambiq & PsiKick)
 - Judicious use of sleep mode



<https://www.elektronikpraxis.vogel.de/vibration-harvester-power-supply-opens-door-to-fix-and-forget-sensor-a-453544/>

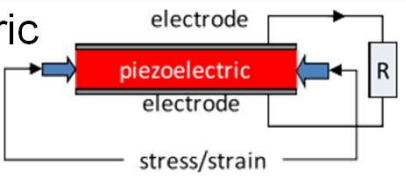
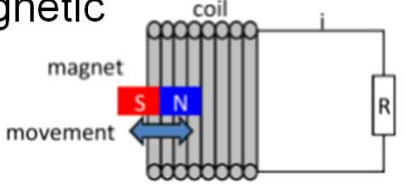
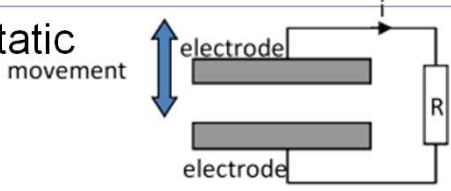
Tatsuakira Masaki et al 2011 J. Micromech. Microeng. 21 104004
<https://www.eejournal.com/article/20140911-imec/>

Comparison to other Vibration Harvesters

	Energy density	Max. Energy density
Piezoelectric 	$W_{max} = \frac{\sigma_y^2 k^2}{2Y}$	$W_{max} = 17.7 \frac{mJ}{cm^3}$
Electromagnetic 	$W_{max} = \frac{B^2}{2\mu_0}$	$W_{max} = 4 \frac{mJ}{cm^3}$
Electrostatic 	$W_{max} = \frac{1}{2} \epsilon_0 E^2$	$W_{max} = 4 \frac{mJ}{cm^3}$

Roundy S. J., 2003, PhD thesis; Boisseau et al., Small-Scale Energy Harvesting, M. Lallart, IntechOpen, 2012; Boland, Justin (2005): Micro Electret Power Generators. California Institute of Technology.

Comparison to other Vibration Harvesters

	Advantages	Disadvantages
Piezoelectric 	• <i>High energy density</i> • <i>High Voltage</i>	• <i>Expensive material</i> • <i>Brittle material</i> • <i>Low current</i>
Electromagnetic 	• <i>Easy to scale up</i> • <i>Magnet can be used as proof mass</i>	• <i>Difficult to miniaturize</i> • <i>Low voltage</i>
Electrostatic 	• <i>Easily miniaturized</i> • <i>Medium to High Voltage</i>	• <i>Low current</i> • <i>Need of voltage source or electret</i> • <i>Typ. high frequency</i>

Roundy S. J., 2003, PhD thesis; Boisseau et al., Small-Scale Energy Harvesting, M. Lallart, IntechOpen, 2012; Boland, Justin (2005): Micro Electret Power Generators. California Institute of Technology.